

Expected Fertility, Labor Market Contracts, and the Gender Wage Gap

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Abstract

This paper studies how employers' expectations about women's future fertility contribute to the gender wage gap. I argue that this expectations-based channel is particularly important in labor markets with persistent employment relationships, where wages are set in advance based on the expected present value of future productivity rather than contemporaneous output. If employers expect women's productivity growth to slow after childbirth, they may discount women's wages before children are born. I test this mechanism using China's 2013 selective relaxation of the One-Child Policy as a natural experiment. Exploiting cross-provincial variation in pre-reform policy enforcement in a difference-in-differences framework, I find that young women's wages declined by 15.3% immediately after the reform, despite no short-run increase in actual fertility. The wage response is concentrated in sectors with more persistent employment relationships, consistent with employers pricing expected future productivity rather than replacement costs. To interpret these findings, I develop a search-and-matching model with wage posting and on-the-job human-capital accumulation, integrated with a household model in which fertility influences future productivity through household time allocation and non-contractible effort. Estimating the model using Chinese labor-market data, I find that gender differences in expected future productivity, arising from the unequal division of household labor, explain nearly the entire pre-reform gender wage gap and approximately 80% of the post-reform widening. These findings suggest that persistent employment relationships can shift part of the child penalty from the post-childbirth period to the wage-setting stage, implying that policies preserving employment alone may be insufficient to eliminate gender wage gaps.

Keywords: gender wage gap; fertility; search and match

JEL codes: J16; J31; J64

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1 Introduction

Despite convergence in educational attainment and rising female labor force participation, sizable gender wage gaps persist across countries, with little evidence of further narrowing in recent decades (Olivetti and Petrongolo, 2016; Blau and Kahn, 2017; Kunze, 2017). A central mechanism behind these gaps is the child penalty: childbirth and early child-rearing often lead women to exit employment, reduce hours, or slow their career progression, thereby depressing cumulative earnings and delaying advancement into managerial positions (Adda, Dustmann and Stevens, 2017; Kleven, Landais and Sogaard, 2019; Goldin, 2021; Cortés and Pan, 2023; Kleven, Landais and Leite-Mariante, 2024).

This paper studies a less explored counterpart to the realized child penalty: an *ex ante* child penalty that arises before children are born. Employers may lower women’s wages not only because they anticipate maternity leave, separation, or replacement costs, but also because they expect future childcare responsibilities to reduce women’s productivity growth after childbirth. This channel can operate even when women remain continuously employed and work the same recorded hours. It is especially relevant in labor markets with persistent employment relationships and rigid wage setting, where current wages reflect expected productivity over a long horizon.

I study this mechanism using the 2013 selective relaxation of China’s One-Child Policy as a natural experiment. The reform allowed a couple to have a second child if either spouse was an only child. It therefore increased the perceived likelihood of a second birth among young women, while actual fertility changed little in the short run. The reform provides a setting in which immediate wage responses can be interpreted as responses to revised fertility expectations rather than to realized childbirth or contemporaneous labor-supply adjustment.

Exposure to the policy varied across provinces. Provinces that had enforced the One-Child Policy more strictly produced larger shares of only children and consequently had larger populations newly eligible for a second birth. I measure historical enforcement intensity using provincial fines for unauthorized births and exploit the interaction between this variation and the timing of the relaxation. Using the China Family Panel Studies, I focus on urban workers aged 18–35 with at least a high-school education, a group more likely to hold salaried

jobs with persistent employment relationships.

The main result is a sharp decline in young women’s wages following the reform. In provinces more exposed to the relaxation, young women’s wages fell by 15.3% relative to those of young women in less-exposed provinces. Comparable young men experienced no significant wage response. A triple-difference specification using men as within-province and within-period controls yields a similar estimate of 14.8%. Employment probabilities did not change significantly for either gender.

Several findings distinguish this effect from a contemporaneous motherhood penalty. Aggregate birth rates did not rise after the first relaxation, and province-year difference-in-differences estimates show no larger increase in births in more-exposed provinces. The wage response is concentrated among women aged 35 or younger and is absent among older women, who face substantially lower expected fertility. These patterns indicate that the immediate wage decline is associated with revised expectations about future fertility rather than realized childbirth or a general reassessment of women’s productivity.

The cross-sectional pattern of the wage response points to the importance of long-term employment relationships. The effect is absent among less-educated workers, who are less likely to hold jobs in which current pay reflects productivity over a long horizon. By contrast, it is substantially larger in government-related employment, including government and Party organizations, public institutions, and state-owned enterprises, where employment is highly persistent and separation risk is low. In these sectors, the estimated wage decline for young women is more than twice the full-sample effect, while the corresponding estimate in non-government sectors is small and statistically insignificant.

This evidence supports a long-horizon expected-productivity channel of statistical discrimination. The mechanism differs from the replacement-cost channel emphasized by [Amano-Patiño, Baron and Xiao \(2020\)](#), in which employers lower women’s wages because they expect childbirth to increase separation and impose hiring or training costs. Here, the penalty arises precisely because women are expected to remain with the employer. When matches are persistent, firms internalize expected productivity over a longer horizon. Lower separation risk can therefore increase, rather than reduce, the extent to which anticipated post-birth productivity differences are reflected in current wages.

To formalize this mechanism, I develop a model that combines household time allocation with wage posting and on-the-job search. In the household block, spouses allocate time among market work, household production, and free time. Employment is discrete: workers either do not work or supply a fixed amount of full-time labor. Conditional on employment, recorded market hours are therefore fixed. Household responsibilities nevertheless reduce free time, including time for sleep, recovery, and career-related activities outside formal working hours. Lower free time reduces effective effort and slows learning-by-doing human-capital accumulation.¹

Because childcare remains unequally distributed, an additional child raises women’s household burden more than men’s. Holding formal employment and market hours fixed, women consequently have less free time, exert lower effective effort, and accumulate human capital more slowly. Fertility therefore affects both current effective productivity and the future growth of productivity, even among women who remain continuously employed full-time.

I embed this household block in a wage-posting model with on-the-job search, building on [Burdett and Mortensen \(1998\)](#). Firms post wages and commit to them for the duration of the match. Workers receive outside offers stochastically and move when a better offer arrives, so wage transitions also capture career advancement. Human capital is match specific and grows at the effort-dependent accumulation rate. Because future effort is not contractible and wages cannot be continuously renegotiated, firms set wages according to the expected present value of productivity over the employment relationship. A downward revision in expected future effort or human-capital growth therefore lowers current posted wages.

The model separates three sources of gender wage differences: household time allocation and its effect on effort and human-capital accumulation, labor-market search frictions such as offer-arrival and separation rates, and employers’ beliefs about future fertility. This decomposition distinguishes wage gaps arising from unequal household responsibilities from those generated by search frictions or changes in fertility expectations.

¹This mechanism is related to [Albanesi and Olivetti \(2009\)](#), who model home hours as increasing the utility cost of market effort. It is also consistent with time-use evidence showing that childcare and household production compress sleep and discretionary time, as well as with the “greedy jobs” mechanism in [Goldin \(2014\)](#), under which continuous availability and off-the-clock investment are rewarded while flexibility is penalized.

I estimate the model using CFPS data from 2012 and 2014, matching gender-specific wage distributions, labor-market transition rates, household-production time by gender and number of children, and observed fertility distributions. I first estimate the pre-reform equilibrium and decompose the 2012 gender wage gap by equalizing, one component at a time, household time allocation, search frictions, and fertility-type distributions across men and women.

The pre-reform estimates indicate that unequal household responsibilities are the dominant source of the baseline gender wage gap. Equalizing the household time-allocation channel across genders eliminates nearly all of the residual wage gap in 2012. By contrast, gender differences in offer-arrival rates, separation rates, and fertility-type distributions contribute relatively little once household time allocation is equalized.

I then estimate the post-reform equilibrium. Because realized fertility and household time allocation changed little immediately after the relaxation, I hold fixed the household time-allocation schedule and human-capital accumulation technology and allow the reform to affect labor-market parameters and employers' beliefs about future fertility. The estimates imply a substantial downward revision in expected future productivity for young women. Applying this belief change to the 2012 labor-market environment accounts for nearly 80% of the simulated 15.5% widening in the gender wage gap. Changes in offer-arrival rates move in the same direction, consistent with greater reluctance to hire or promote young women, but explain only a small portion of the widening.

Taken together, the reduced-form and structural evidence suggest that persistent employment relationships can bring the child penalty forward. When wages are fixed over long horizons, anticipated post-birth productivity differences are capitalized into wages before childbirth occurs. Policies that preserve women's employment after childbirth, such as maternity-leave mandates or employment-protection rules, may therefore reduce separations without eliminating the wage penalty. By extending the horizon over which firms internalize expected productivity differences, such policies may shift part of the penalty from the post-birth period to the earlier hiring or wage-setting stage.

This argument does not imply that maternity leave or employment protection is undesirable. Rather, it suggests that position-preserving policies alone are unlikely to eliminate gender wage gaps unless they also address the underlying asymmetry in household responsibilities.

Policies that reduce childcare burdens or increase fathers’ involvement directly weaken the source of expectation-based pricing.

Although the empirical setting is China and the outcome is the gender wage gap, the mechanism extends to labor markets characterized by persistent matches and rigid pay. These include government and state-affiliated employment and many high-skill occupations in which long employment horizons arise from costly matching, firm-specific training, or expensive replacement. In such settings, employer beliefs about future effort and productivity can have large effects on current wages. The same logic may apply to expectations related not only to fertility, but also to immigration constraints, health risks, or other anticipated events that affect future productivity.

After a brief discussion of the literature, the remainder of this paper is organized as follows. Section 2 presents the empirical setting, background on China’s One-Child Policy, the identification strategy, data, and main results, together with placebo and mechanism tests. Section 3 develops the theoretical framework that links household time allocation, effort, and learning-by-doing to wage posting with on-the-job search, and characterizes equilibrium. Section 4 maps the model to the data: it details the numerical solution, reports the estimation for 2012 and 2014, and conducts counterfactual decompositions. Section 5 concludes and discusses policy implications. Proofs and technical details, together with relevant discussions, are gathered in a final Appendix.

Literature Review This paper speaks to several strands of research. First, a large literature documents persistent gender wage gaps and traces a central role to children. Surveys show that gaps have narrowed only slowly despite convergence in education and labor force participation (Blau and Kahn, 2017; Olivetti and Petrongolo, 2016; Kunze, 2017), with childbirth generating sizable, long-lived penalties through reduced hours and interrupted human capital accumulation (Adda, Dustmann and Stevens, 2017; Kleven, Landais and Søgaaard, 2019; Cortés and Pan, 2023; Goldin, 2021; Kleven, Landais and Leite-Mariante, 2024). Classic models of household specialization and effort (Becker, 1985; Albanesi and Olivetti, 2009; Albanesi, Olivetti and Petrongolo, 2023) and the “greedy jobs” perspective

(Goldin, 2014) rationalize why home responsibilities compress focus, availability, and off-the-clock learning that sustain productivity growth at work. Correll, Benard and Paik (2007) applies a laboratory experiment to causally identify the penalization of mothers in hiring and wages, reinforcing the idea that employers perceive women as less productive due to their family responsibilities. However, while most of this literature focuses on post-birth employment outcomes, my paper emphasizes that gender-based disadvantages emerge even before childbirth and for all women regardless of their fertility intentions, particularly in less mobile labor markets where employers preemptively adjust wages based on expected, rather than actual, fertility behavior.

A second strand links gender gaps to statistical discrimination in frictional labor markets. Structural and reduced-form studies emphasize higher expected separations and replacement costs for mothers (Bowlus and Grogan, 2009; Bartolucci, 2013; Amano-Patiño, Baron and Xiao, 2020). The contribution of my paper is to shift the mechanism from expected separations to expected productivity while separation is negligible: when pay is set against the present value of match productivity, beliefs about future effort and learning-by-doing receive more weight, so markdowns can arise even without separations, and, in fact, are larger when separations are rarer and matches are longer.

Third, the paper contributes to work on China’s fertility policy and women’s labor outcomes. The One-Child Policy (OCP) and its local enforcement have been studied extensively (Gu et al., 2007; Scharping, 2013; Ebenstein, 2010; Huang, Lei and Zhao, 2016), and recent papers exploit relaxations to study realized fertility and labor effects, including field-experimental evidence on employer discrimination (He, Li and Han, 2023), city-level impacts on gaps (Agarwal et al., 2024), and post-birth outcomes (Li, 2022). In contrast, I use nationally representative survey data to capture aggregate effects and leverage the 2013 selective relaxation as a shock to expected fertility to isolate forward-looking wage setting under contracts. Rather than simply documenting a wage decline, I develop a model in which effort drives human capital growth and estimate it before and after the reform, quantifying how the unequal division of household labor is translated into expectation-driven pricing and how much of the post-reform widening it can explain.

Finally, this paper complements a growing literature showing that well-intentioned

maternity-leave and anti-discrimination laws may inadvertently worsen women’s labor market outcomes. Using detailed job-platform data from China, [Fang, Hu and Yu \(2025\)](#) show that after the 2021 extension of statutory maternity leave by 10–50 days (roughly 22%), callback rates for young women fell by about 17%, indicating heightened hiring discrimination. [Thomas \(2020\)](#) develops a two-period model that predicts and empirically confirms that maternity leave provisions can lower women’s promotion probabilities by pooling women with heterogeneous fertility preferences into the same candidate pool. Similarly, [Di Paolo and Marcolin \(2025\)](#) studies the passage of the 1978 U.S. Pregnancy Discrimination Act, which prohibited employers from discriminating on the basis of pregnancy, and finds that the reform reduced women’s employment rates, suggesting that firms substituted away from hiring women of childbearing age. In the Indian context, [Gupta \(2023\)](#) document that extensions of paid maternity leave led firms to replace female employees with male counterparts.

This paper differs from these studies along two key dimensions. First, much of the literature identifies “child-penalty” effects on the extensive margin—employment, hours, or promotion—and finds limited or mixed wage effects. In contrast, in China’s high-skilled, contract-intensive labor market, I document a sizeable decline in young women’s wages after the 2013 fertility-policy relaxation, with no corresponding drop in employment. Two features plausibly underpin this pattern: (i) in a rapidly growing economy, outright non-employment among qualified women is rare, so adjustment occurs via lower starting offers rather than job loss; and (ii) although “equal pay for equal work” is formally mandated, enforcement is incomplete ([World Bank, 2024](#)), leaving room for pre-hire wage setting and job assignment to serve as the margin of statistical discrimination. These findings underscore that regulating pay equality alone cannot eliminate statistical discrimination. When wage floors constrain within-position differentiation, employers may shift discrimination to the hiring margin—reducing offers, limiting advancement, or avoiding female candidates altogether. Second, the studies mentioned above identify expected fertility effects through changes in maternity leave duration or the frequency of pregnancy-related episodes. However, the economic impact of another child should extend far beyond the childbirth period and encompass the entire child-rearing phase. The relaxation of China’s One-Child Policy introduces an exogenous shift in the expected number of children while keeping maternity leave and other labor market conditions

stable. Therefore, the exogenous shift used in this paper allows for a more comprehensive capture of the expected fertility effect on women’s labor market outcomes, accounting for the cumulative impact of childcare and household responsibilities over time rather than the short, episodic disruption surrounding childbirth.

2 Empirical Evidence

This section exploits the 2013 selective relaxation of China’s One-Child Policy (OCP) to estimate how an increase in expected fertility affects young women’s wages. The empirical design combines three sources of variation: the timing of the policy relaxation, cross-province differences in pre-reform enforcement intensity, and gender. I first estimate difference-in-differences specifications separately by gender and then use men as an additional control group in triple-difference specifications.

The main result is a 15.3% decline in young women’s wages in provinces more exposed to the policy relaxation. In contrast, neither men’s wages nor employment rate change significantly. I also find no contemporaneous increase in birth rates, either in the aggregate time series or in province-year regressions. These patterns suggest that the wage decline reflects a response to higher expected fertility rather than an immediate adjustment in fertility or labor supply.

Additional evidence points to the importance of long-term employment relationships. The wage response is concentrated among younger women, is absent among less-educated workers, and is substantially larger in government-related employment, where employment relationships tend to be especially persistent. These findings motivate the theoretical framework developed in the next section, in which firms price expected future productivity into current wages.

The remainder of this section proceeds as follows. Section 2.1 describes the One-Child Policy and its selective relaxation. Section 2.2 presents the empirical design. I then describe the data and sample, report the main results, and examine evidence on the expected-fertility mechanism.

2.1 Background: The One-Child Policy and Its Relaxation

China’s modern family-planning campaign began in 1971 and was initially implemented through voluntary targets and administrative encouragement. In 1979, the government introduced a substantially stricter population-control regime, commonly known as the One-Child Policy. Urban couples were generally limited to one child, while many rural couples were allowed a second child if their first child was a girl. The policy was enforced through financial penalties and administrative sanctions, including wage deductions for salaried workers, reduced rural land allocations, restrictions on access to public services, and fines for unauthorized births. Fertility fell sharply during this period. According to data summarized by the United Nations, China’s total fertility rate declined from 2.9 in 1979 to 1.6 by 2000 and remained low thereafter, as shown in Figure 1.

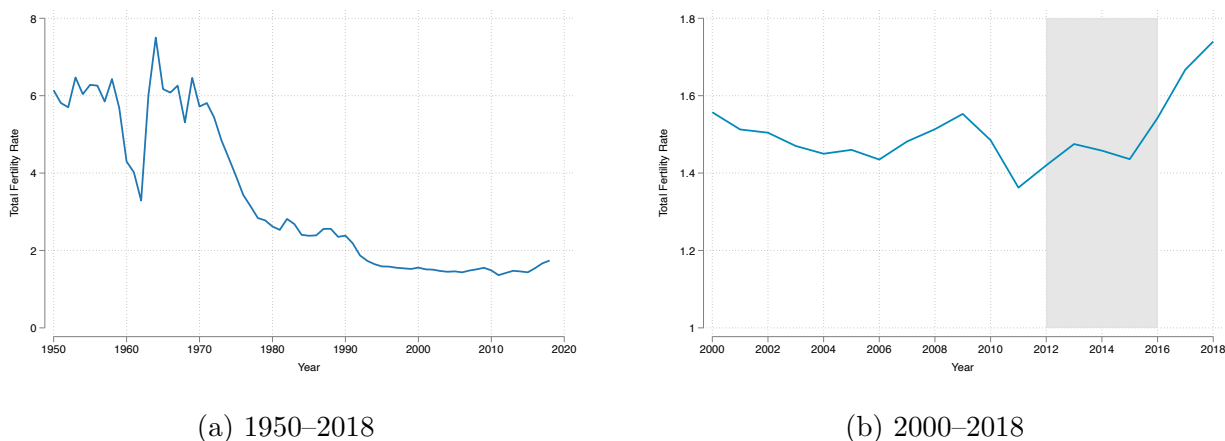


Figure 1: Trends in the total fertility rate in China. The shaded region in (b) denotes the sample period. *Sources:* United Nations, Department of Economic and Social Affairs, Population Division (2019).

Concerns about persistently low fertility and population aging later prompted a consecutive relaxation of the OCP. The first major relaxation, known as the “selective two-child” policy, was announced in 2013 and allowed a couple to have a second child if either spouse was an only child. Provincial implementation occurred primarily in 2014 and 2015. Despite widespread expectations of a baby boom, birth rates responded little in the short run, as shown in Figure 3a.² The central government subsequently introduced a universal two-child

²The increase in births around 2012 may partly reflect the Year of the Dragon. In Chinese zodiac beliefs, the Dragon is considered especially auspicious, and some parents time childbirth accordingly. Dragon-year fertility effects have been documented in Chinese populations (Goodkind, 1991; Agarwal et al., 2021).

policy, announced in November 2015 and effective in January 2016, followed by a three-child policy in 2021.

Measuring OCP enforcement intensity. Although the OCP applied nationwide, enforcement intensity varied substantially across provinces. This variation reflected differences in local regulations, administrative capacity, and social norms regarding fertility. The main enforcement instrument was the fine for unauthorized births, often referred to as a “social compensation fee.” These fines were substantial and varied widely across provinces. For example, Zhejiang, Fujian, and Guangdong—three economically developed provinces along China’s southeastern coast—imposed different fine schedules. Under Zhejiang’s family-planning regulations, couples who exceeded birth quotas were required to pay 20% to 50% of household income annually for five years. In Fujian, fines were around two to three times the household’s income in the previous year. In Guangdong, the 2002 regulations specified fines between three and six times annual disposable income. In addition to monetary penalties, urban residents employed in government agencies or state-owned enterprises could face job loss or reductions in welfare benefits for violating birth quotas. In some regions, coercive enforcement measures were also reported.

To obtain a comparable measure of enforcement intensity, I use province-level fine levels expressed as multiples of average annual household income. [Scharping \(2013\)](#) compiles official provincial fine schedules for 1979–2000. Building on these schedules, [Ebenstein \(2010\)](#) combine official regulations with China Health and Nutrition Survey data to construct annual province-specific fine rates. These measures have been widely used as proxies for OCP enforcement intensity (e.g., [Ebenstein, 2010](#); [Huang, Lei and Zhao, 2016](#)).

My baseline measure, denoted by *FinesRate*, is the 2000 cross-section of these province-level fine rates. A higher value indicates stricter pre-reform enforcement and a larger expected penalty for an unauthorized birth. [Figure 2](#) displays the cross-provincial variation in *FinesRate* and its relationship with provincial gross regional product.

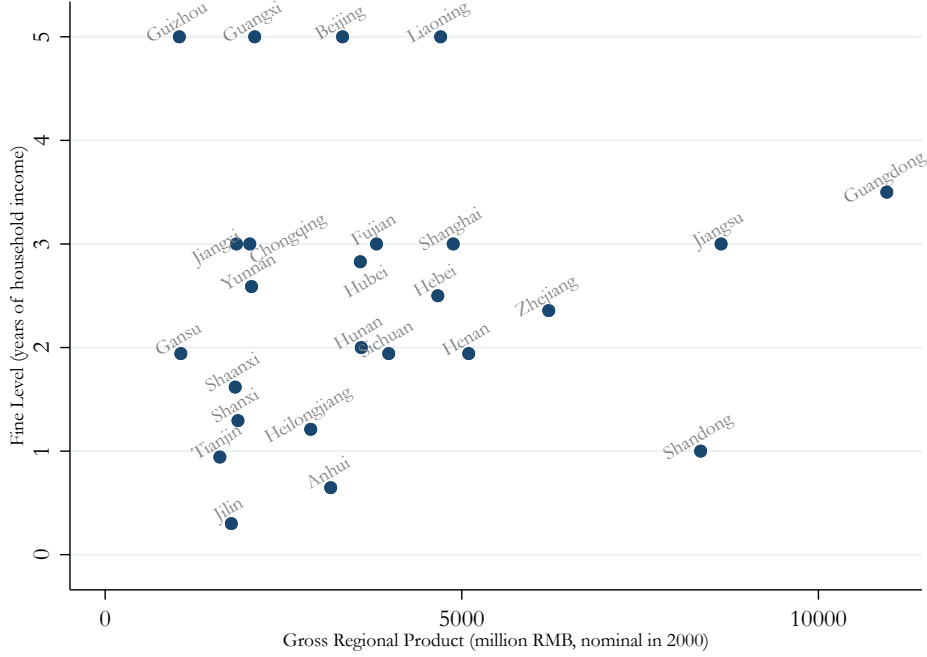


Figure 2: Harmonized One-Child Policy fine rates by province, 2000. Fine rates are expressed as multiples of average annual household income. *Sources:* Fine rates are from [Ebenstein \(2010\)](#), based on [Scharping \(2013\)](#) and the China Health and Nutrition Survey; provincial gross regional product is from the *National Bureau of Statistics of China*.

2.2 Empirical Strategy

The selective relaxation provides a natural experiment for studying how expected fertility affects young women’s wages. I exploit cross-province variation in exposure to the reform using a difference-in-differences design.

The selective two-child policy applied only to couples in which at least one spouse was an only child. Provinces with stricter historical enforcement produced larger shares of only children and therefore had larger populations newly eligible for a second birth. The reform consequently generated a larger increase in expected fertility in provinces with higher values of *FinesRate*. I therefore use historical enforcement intensity as a continuous measure of exposure to the policy relaxation.

To validate this interpretation, I examine whether respondents born in provinces with higher *FinesRate* were more likely to be only children before the reform. I use the 2010 CFPS adult questionnaire, which records respondents’ numbers of siblings, including deceased siblings. Enforcement intensity is assigned according to province of birth rather than

current province of residence. The sample is restricted to respondents aged 18–30 in 2010, corresponding approximately to cohorts born after the introduction of the OCP. The primary validation sample further restricts respondents to those currently living in urban communities, consistent with the main wage analysis.

Table 1: Only-Child Share by High- and Low-Fine Birth Provinces

Sample	Low fine	High fine	Difference
All current-urban age 18–30	0.379	0.497	0.118
Urban/high-edu age 18–30	0.455	0.624	0.169
Wage-regression-like age 18–30	0.428	0.630	0.202

Notes: The table reports the share of respondents with no siblings in the 2010 CFPS adult questionnaire. High-fine provinces are birth provinces with 2000 fine rates at or above 2.516 multiples of average annual household income. The all-current-urban sample includes respondents ages 18–30 currently living in urban communities. The main-like sample additionally restricts to at least 12 years of schooling. The wage-regression-like sample further restricts to employed respondents with positive wage income.

Table 1 supports the proposed measure of reform exposure. Across all three samples, respondents born in high-fine provinces are substantially more likely to be only children. The difference ranges from 11.8 to 20.2 percentage points and is largest in the sample that most closely resembles the main wage-regression sample. This evidence is consistent with stricter historical enforcement producing a larger population eligible for a second birth under the selective relaxation.

The empirical design also requires that historical enforcement intensity not proxy for provincial economic conditions that independently affect wages. To assess this concern, I regress *FinesRate* on provincial GDP, urban unemployment, average urban wages, and urban employment using data from the National Bureau of Statistics of China.³ Because GDP and unemployment are available in 2000, whereas wage and employment measures are available only from 2008 onward, I estimate one cross-sectional specification using 2000 GDP and unemployment and another using all four variables measured in 2013.

Table 2 shows small and statistically insignificant relationships between *FinesRate* and provincial economic conditions. Joint F-tests also fail to reject the hypothesis that the

³The analysis focuses on the urban sector because formal wage employment, the main outcome of interest, is concentrated in urban China.

Table 2: Fine Rates are Unrelated to Provincial Economic Conditions

	(1) year = 2000	(2) year = 2013
GDP	0.000 (0.000)	-0.000 (0.000)
Unemployment rate	-0.349 (0.430)	-0.585 (0.509)
Number of employed		0.002 (0.004)
Average wage		0.000 (0.000)
Constant	3.477** (1.429)	3.647 (2.635)
Observations	25.000	25.000
R^2	0.034	0.166
F-statistic	0.387	0.998
P-value	0.683	0.432

Notes: Standard errors are reported in parentheses. This table examines whether provincial enforcement intensity of the One-Child Policy (*FinesRate*) is correlated with economic conditions across provinces. Column (1) uses cross-sectional data for 2000, including provincial GDP and urban unemployment rates; Column (2) uses 2013 data, adding average urban wages and total urban employment. Data on economic variables come from the *National Bureau of Statistics of China*, and *FinesRate* is constructed from [Ebenstein \(2010\)](#).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

included covariates have no explanatory power for *FinesRate*, with p-values of 0.683 and 0.432. I therefore find no evidence that historical enforcement intensity is systematically related to the observed provincial economic conditions.

Difference-in-differences specification. I follow a standard difference-in-differences strategy, and the estimation equation is

$$\log \text{wage}_{it} = \beta_0 + \beta_1 \text{yschool}_{it} + \beta_2 \text{age}_{it} + \beta_3 \text{age}_{it}^2 + \beta_4 \text{Treat}_{pt} + \mu_p + \gamma_t + \varepsilon_{it} \quad (1)$$

where i refers to individual worker, t calendar time, and p provinces. Using the logarithm level of nominal yearly salary income, “log wage _{it} ”, as the dependent variable, I include years of schooling, “yschool”, “age”, and “squared age” as the independent variables based on the Mincer equation. Province fixed effects (μ_p) and year fixed effects (γ_t) are also included in this specification. The coefficient of interest is β_4 in front of the binary indicator Treat_{pt} , which is constructed by the time indicating variable Time_t and treatment group indicating variable FinesHigh_p :

$$\text{Treat}_{pt} = \text{FinesHigh}_p \times \text{Time}_t$$

where

$$\text{Time}_t = \begin{cases} 1, & \text{if } t > 2013 \\ 0, & \text{if } t \leq 2013 \end{cases}, \text{ and } \text{FinesHigh}_p = \begin{cases} 1, & \text{if } \text{FinesRate}_p > 2.5 \\ 0, & \text{if } \text{FinesRate}_p \leq 2.5 \end{cases}$$

where 2.5 is the median of fines rates for all provinces in 2000 from [Ebenstein \(2010\)](#), in terms of *years of household income*. In other words, I assign provinces that had a higher fine rate for excess births in 2000 to the treatment group and others to the control group. $\text{Treat}_{pt} = 1$ implies workers in province p are under treated in year t , in comparison with the same-gender workers that are either never treated or not yet treated. As the fines rates are continuously measured, I create a new variable to capture treatment intensity, denoted as $\text{Con_Treat}_{pt} = \text{FinesRate}_p \times \text{Time}_t$, and substitute for the binary Treat_{pt} variable in regression model described by Equation 1.

I apply Equation 1 to female workers and male workers to assess differential post-policy effects by gender. As a complementary approach, I also use men’s wages as the natural control group for possible wage-determining factors that are systematically different across the treated and untreated province-year combinations, and following [Jayachandran and Lleras-Muney \(2009\)](#) the estimating model for Triple DiDs is

$$\begin{aligned} \log \text{wage}_{it} = & \beta_0 + \beta_1 \text{yschool}_{it} + \beta_2 \text{age}_{it} + \beta_3 \text{age}_{it}^2 \\ & + \beta_4 \text{Treat}_{pt} \times \text{female}_i + \mu_p \times \gamma_t + \gamma_t \times \text{female}_i + \mu_p \times \text{female}_i + \varepsilon_{it} \end{aligned} \quad (2)$$

The model incorporates a full set of double interactions of fixed effects, including the province-year, gender-year, and province-gender fixed effects. This enables us to interpret the coefficient of interest, β_4 , as the impact of an increase in expected fertility rates on female wages, accounting for all variations in province, year, and gender. I use the continuously measured treatment intensity `Con_Treat` as a second treatment specification in the estimation as well.

2.3 Data and Sample Selection

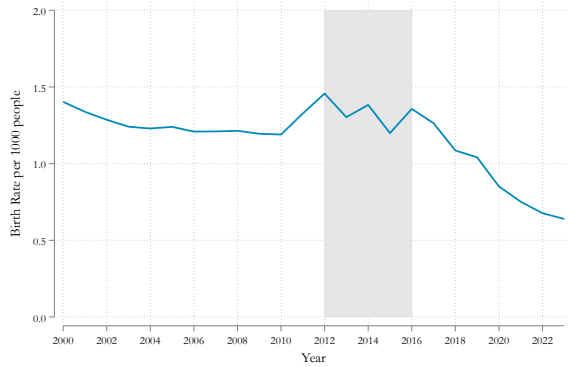
The primary dataset is the China Family Panel Studies, a nationally representative household survey launched in 2010 and conducted biennially thereafter. I use the 2012, 2014, and 2016 survey waves. I exclude 2010 because changes in the employment questionnaire make employment measures difficult to compare with later waves.⁴ I end the sample in 2016 because the universal two-child policy took effect that year, changing the policy environment relevant to fertility expectations.

The CFPS covers 25 provinces. It excludes several minority-concentrated regions, including Xinjiang, Tibet, Qinghai, Inner Mongolia, Ningxia, and Hainan. A small number of respondents in later waves report living in these provinces following migration. I exclude these observations because of their small sample size and because the original OCP rules differed for many minority populations.

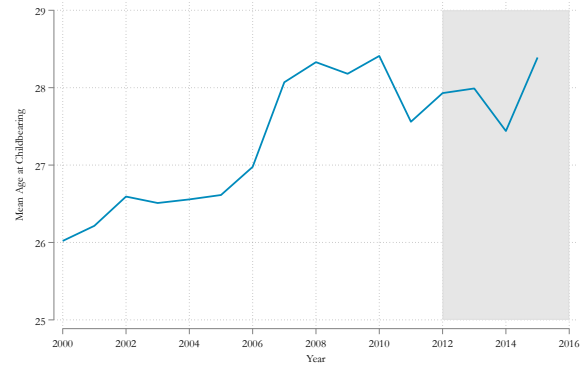
The main sample consists of urban residents aged 18–35 with at least 12 years of schooling. The age restriction focuses the analysis on women in their prime childbearing years. The schooling restriction concentrates the analysis on workers more likely to hold salaried jobs and to form persistent employment relationships in which current wages may depend on expected future productivity. Section 2.5.2 examines less-educated workers separately.

Table 3 reports summary statistics for respondents satisfying the sample restrictions and reporting nonmissing wage income. Sample sizes are broadly similar across survey waves, and average age ranges from 27.8 to 29.2. The share holding urban *hukou* declines from approximately 85% in 2012 to 71–74% in 2016, while the minority share rises from 2–3% to

⁴The measured employment rate is 62.96% in 2010, compared with 83.29%, 84.82%, and 85.06% in 2012, 2014, and 2016.



(a) Birth rate in China



(b) Mean age at childbearing in China

Figure 3: Fertility-related statistics in China since 2000. The shaded region denotes the sample period. *Sources:* United Nations, Department of Economic and Social Affairs, Population Division (2019).

Table 3: Summary Statistics by Year and Gender

	2012		2014		2016	
	Female	Male	Female	Male	Female	Male
Sample size	466	486	540	563	506	556
Age	28.34	28.74	27.81	28.45	28.68	29.15
Urban <i>hukou</i> (%)	85.16	84.77	75.10	78.03	71.34	73.92
Minority (%)	3.27	2.34	6.31	4.24	6.36	4.76
Years of schooling	14.37	14.26	14.59	14.55	14.72	14.69
	(1.748)	(1.806)	(1.829)	(1.877)	(1.763)	(1.947)
ln(wage)	9.99	10.23	10.12	10.43	10.20	10.53
	(0.916)	(0.921)	(0.915)	(0.980)	(1.271)	(1.114)
Total <i>n</i>	952		1,103		1,062	
Gender wage ratio (F/M)	0.7890		0.7349		0.7178	

Notes: Entries are means; standard deviations in parentheses. Urban *hukou* and minority are binary variables, whose means are displayed in percent. The gender wage ratio is computed as $\exp(\ln w_F - \ln w_M)$, a geometric-mean wage ratio used to reduce sensitivity to extreme wage values.

approximately 5–6%. Average schooling and mean log wages increase modestly across waves. The geometric-mean female-to-male wage ratio declines from 78.90% in 2012 to 73.49% in 2014 and 71.78% in 2016, indicating a widening unadjusted gender wage gap during the sample period.

2.4 Empirical Results

Table 4 reports estimates of Equations (1) and (2). Standard errors are clustered at the province level, which is the level of treatment variation.

Table 4: Effect of OCP Relaxation on Female/Male Workers' Wages

	(1) Female	(2) Male	(3) Female	(4) Male	(5) Triple-DiD	(6)
Dependent variable: log wage						
Treat	-0.153*	-0.031				
	(0.083)	(0.084)				
Treat (cont.)			-0.059*	0.006		
			(0.031)	(0.025)		
Treat $\times$ female					-0.148*	
					(0.078)	
Treat (cont.) $\times$ female						-0.075**
						(0.030)
Observations	1467	1519	1467	1519	2986	2986
R^2	0.200	0.216	0.200	0.215	0.239	0.239

Notes: Standard errors in parentheses. “Treat” is a binary indicator for exposure to the OCP relaxation; “Treat (cont.)” is a continuous exposure measure. Triple-DiD columns report the interaction with the female indicator. The table reports coefficients of interest; all specifications control for years of schooling, age, and age squared. The full regression table is reported in Appendix A.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The results for Equation 1 with binary indicator variable, Treat_{pt} , are presented in the first two columns. The estimated coefficient for Treat_{pt} is negative for female workers, indicating a statistically significant decrease of 15.3 percent in wages for young women in provinces experiencing more exposure to the policy change, as compared to unaffected females. However,

for male workers, the estimated coefficient is not statistically significant from 0, suggesting that the change in policy does not impact their wages.

Columns (3) and (4) exhibit the results using continuously measured treatment variable Con_Treat_{pt} . Notably, the alteration in treatment measurement does not alter the primary finding: when assessed based on fines rates, female workers in provinces that previously imposed fines equivalent to 1 more year of household income for excess births experience a 5.9 percent decrease in wages after OCP relaxation, while if we multiply this number by the median 2.5, the result is 14.75 percent, quite comparable to what we got from binary treatment indicator. Conversely, the strictness of OCP implementation in the past did not have a significant effect on the wages of male workers.

In addition to the key coefficients of variable Treat_{pt} and Con_Treat_{pt} , other estimated results fall within the range found in the literature. Specifically, an additional year of schooling is associated with an 8.9 percent increase in wages for women and a 6.7 percent increase for men. Furthermore, wages exhibit a positive correlation with age (experience), although with a significantly diminishing return.

Triple difference-in-differences results specified in Equation 2 are presented in the last two columns. The effects of increased expected fertility rates on young females' wages are negative and significant. Under a counterfactual scenario without policy relaxation, the wage level for the same female worker would have risen by 14.8 percent. Taking the average real wage level before the policy shift for females in the treatment group as a reference point, this implies an annual wage income loss of approximately $34,650.87 \times 0.148 = 5,128.33$ RMB, which is approximately 833.87 USD using the 2013 exchange rate. If we interpret the results by the continuously-measured treatment intensity, it implies that if the province where a woman lives increased its fine by one year of household income for an excess birth, the female's annual wage income would decrease by 7.5 percent, which equals to 2598.82 RMB and 422.57 USD.

Employment responses. When it comes to wage change, the naturally-related question is whether the employment rates change simultaneously. If the possibilities of finding jobs for young females increase after the policy relaxation, the wage decrease could happen because

Table 5: Effect of OCP Relaxation on Female Workers' Employment Status

	Panel A: Binary Treatment			Panel B: Cont. Treatment		
	Female	Male	Triple-DiD	Female	Male	Triple-DiD
Dependent variable: waged employment						
Treat	0.115 (0.125)	-0.030 (0.140)				
Treat $\times$ female			0.147 (0.220)			
Treat (cont.)				0.026 (0.049)	0.063 (0.042)	
Treat (cont.) $\times$ female						-0.035 (0.075)
Observations	2241	2068	4296	2241	2068	4296

Notes: Standard errors in parentheses. “Treat” is an indicator for exposure to the OCP relaxation; “Treat (cont.)” is a continuous exposure measure. “Triple-DiD” includes the interaction with the female indicator. The table reports coefficients of interest; all specifications control for years of schooling, age, and age squared. The full regression table is reported in Appendix A.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

more low-skilled people are in the labor market pool and the average wage level is lower. To test this hypothesis, I run the same regression equations as in Section 2.2 but now use “*have a waged job or not*” as the dependent variable and, therefore, apply a Probit model.

The results are shown in Table 5. None of the key coefficients are significant, implying that the employment rates for either males or females do not respond to the policy relaxation. The selection criteria for the labor market do not change for young females, but the payments for working decrease.

2.5 Evidence on the Expected Fertility Mechanism

2.5.1 Alternative Explanations

The baseline estimates show that wages decline for young women following the selective relaxation of the One-Child Policy. This subsection examines whether the estimated effect is indeed driven by changes in expected fertility. I consider two alternative explanations.

First, the observed wage decline could reflect an increase in actual births and the associated motherhood penalty. Second, employers may have revised their assessment of women’s productivity more broadly, rather than specifically responding to expected fertility among women of childbearing age. The evidence below provides little support for either explanation.

Actual vs. Anticipated Fertility Increases. A natural alternative explanation is that the widening gender wage gap reflects an increase in actual births rather than changes in expected fertility. Under the motherhood-penalty channel, women may earn less because childbirth reduces labor supply, increases the demand for workplace flexibility, or induces transitions into lower-paying jobs (Correll, Benard and Paik, 2007). If this mechanism drives the results, birth rates should increase following the policy relaxation, particularly in provinces with greater policy exposure.

To examine this prediction, I first use province-year birth rates (live births per 1,000 population) from the National Bureau of Statistics of China for the same 25 provinces as in the main analysis over 2011–2019. Figure 3 plots the aggregate trend. Birth rates peak in 2012, remain broadly stable around the 2013 policy relaxation, and decline sharply after 2016. Thus, within the sample period (2012–2016), there is no visible increase in births following the policy change.

I next estimate province-year difference-in-differences specifications analogous to those in Section 2.2. For these regressions, I express the birth rate in percent, equivalent to dividing the births-per-1,000 measure by 10. Let Time_t denote the post-2013 period, FinesHigh_p indicate provinces with above-median 2000 fine rates, and FinesRate_p denote the continuous measure of enforcement intensity. The binary treatment is defined as $\text{Treat}_{p,t} = \text{Time}_t \times \text{FinesHigh}_p$, while the continuous treatment is $\text{ConTreat}_{p,t} = \text{Time}_t \times \text{FinesRate}_p$. The baseline specification is

$$\text{birth_rate}_{p,t} = \beta_0 + \beta_1 \text{Treat}_{p,t} + \mu_p + \gamma_t + \varepsilon_{p,t},$$

with an analogous specification replacing $\text{Treat}_{p,t}$ by $\text{ConTreat}_{p,t}$. Columns (1)–(2) of Table 6 report pooled estimates without fixed effects, while Columns (3)–(4) include province and year fixed effects. Standard errors are robust.

Across all specifications, the treatment effects are economically small and statistically

Table 6: Impact of OCP Relaxation on Birth Rates

	(1) Binary	(2) Continuous	(3) Binary	(4) Continuous
Dependent variable: birth rate (%)				
Time=1 $\times$ Treat=1	0.030 (0.072)			
Time=1 $\times$ fines_2000		0.003 (0.034)		
Treat			0.030 (0.021)	
Treat (cont.)				0.003 (0.008)
Observations	216	216	216	216
R^2	0.011	0.018	0.928	0.928

Notes: Province-year panel for 25 provinces, 2011–2019. The regression outcome is the birth rate expressed in percent, equivalent to the published births-per-1,000 measure divided by 10; coefficients are therefore in percentage points. Standard errors are robust. *Time*= 1 for post-2013 years; *FinesHigh*= 1 for provinces with above-median 2000 OCP fine rates; *FinesRate* is the continuous fine measure. The table reports coefficients of interest; the full regression table is reported in Appendix A.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

insignificant, indicating no differential birth-rate response in provinces more exposed to the policy relaxation. Taken together, the aggregate time series and the province-year regressions provide little evidence that actual fertility increased following the reform. The observed wage decline therefore appears to reflect changes in expected, rather than realized, fertility.

Senior Women as a Placebo Group. A second alternative explanation is that the policy relaxation led employers to revise their assessment of women’s productivity more generally, rather than through expected fertility. If so, the wage response should also be observed among women who had largely passed their childbearing years.

I therefore use women older than 35 as a placebo group. Relative to younger women, these workers face substantially lower expected fertility and should be much less affected by the policy relaxation under the proposed mechanism. I first estimate the continuous-treatment difference-in-differences specification separately for senior women. I then estimate a female-

only triple-difference specification that interacts the post-relaxation treatment intensity with an indicator for being age 35 or younger.

Table 7: Senior Women as an Additional Control Group

	Senior female (1)	Young female (2)	Triple-DiD (3)
Dependent variable: log wage			
Treat (cont.)	-0.019 (0.047)	-0.059* (0.031)	
Treat (cont.) $\times$ young			-0.047* (0.025)
Observations	1252	1467	2719
R^2	0.201	0.200	0.207

Notes: The dependent variable is log wage. The sample is restricted to urban, employed women with at least 12 years of schooling and positive wages. Senior female workers are older than 35; young female workers are age 35 or below. The triple-difference column compares young women with senior women. All specifications control for years of schooling, age, and age squared. Columns (1)–(2) include year and province fixed effects. Column (3) includes province-by-year fixed effects, year fixed effects, and province-by-age-group fixed effects. Standard errors are clustered at the province level. The table reports coefficients of interest; the full regression table is reported in Appendix A.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7 shows that the estimated wage decline remains concentrated among younger women. The triple-difference estimate is negative and statistically significant, indicating that the policy had a substantially larger effect on women who remained exposed to fertility risk. These findings reinforce the interpretation that the estimated wage response is driven by changes in expected fertility among women of childbearing age, rather than by a contemporaneous reassessment of women’s productivity more broadly.

Having ruled out these alternative explanations, I next examine whether the cross-sectional pattern of the estimated effects is consistent with the proposed mechanism.

2.5.2 Evidence on Long-Term Employment Relationships

The evidence so far points to a mechanism in which firms incorporate expected future productivity into current wage offers. This mechanism should be strongest in jobs characterized

by long-term employment relationships, where wages are more likely to reflect the expected present value of future productivity. By contrast, if employment relationships are relatively short or labor markets are more flexible, firms can adjust employment or wages after productivity is realized, reducing the need to price expected fertility into current wages.

Less-Educated Workers. To examine this prediction, I repeat the baseline analysis using urban workers whose education is below high school completion. Relative to the main sample, these workers are less likely to work in jobs characterized by long-term employment relationships and are therefore less likely to have wages determined by expected future productivity.

Table 8 shows little evidence of a wage decline among less-educated women following the policy relaxation. This finding is consistent with the expected-fertility mechanism: when employment relationships are shorter, expected future productivity is less likely to be capitalized into current wages.

Government-Related Employment. The same logic implies that the wage response should be strongest in labor-market segments where long-term employment relationships are particularly prevalent. Government-related employment—including government and Party organizations, public institutions, research institutes, and state-owned or state-controlled enterprises—provides such an environment.

I therefore split the sample into government-related and non-government sectors and re-estimate the female difference-in-differences specification in Equation (1), together with the triple-difference specification in Equation (2). Results are reported for both the binary and continuous treatment measures.

Table 9 shows that, despite the smaller sample size, the wage response is strongest in government-related employment. Under the continuous treatment measure, young women’s wages decline significantly, with point estimates larger than those in the full sample (cf. Table 4). In contrast, estimated wage changes in non-government sectors are small and statistically insignificant. Taken together, the evidence from less-educated workers and government-related employment consistently points to long-term employment relationships

Table 8: Effect of OCP Relaxation on Wages, Low-Education Sample

	Panel A: Binary Treatment			Panel B: Cont. Treatment		
	Female	Male	Triple-DiD	Female	Male	Triple-DiD
Dependent variable: log wage						
Treat	-0.067 (0.246)	0.171 (0.147)				
Treat $\times$ female			-0.264 (0.251)			
Treat (cont.)				-0.034 (0.115)	0.119*** (0.037)	
Treat (cont.) $\times$ female						-0.140 (0.094)
Observations	511	596	1107	511	596	1107
R^2	0.158	0.142	0.245	0.158	0.146	0.246

Notes: Standard errors in parentheses. The sample includes urban workers ages 18–35 with fewer than 12 years of schooling. “Treat” is a binary indicator for exposure to the OCP relaxation; “Treat (cont.)” is a continuous exposure measure. Triple-DiD columns report the interaction with the female indicator. The table reports coefficients of interest; all specifications control for years of schooling, age, and age squared. The full regression table is reported in Appendix A.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

as the environment in which expected fertility is most strongly capitalized into wages.

Table 10 further illustrates the persistence of employment relationships in China’s high-skilled labor market. Between 2010 and 2016, 67.9% of workers remained with the same employer two years later, compared with approximately 40.5% in the United States over a similar horizon. More than half of respondents report that their current job is also their first job, with the share particularly high among tertiary-educated workers. Government-related employment further reinforces this persistence through highly stable employment relationships.

Table 9: Wage Regressions by Government-Related Sector

Panel A: Government-related				
	Female		Triple-DiD	
	(1)	(2)	(3)	(4)
Dependent variable: log wage				
Treat	-0.191			
	(0.151)			
Treat (cont.)		-0.138***		
		(0.045)		
Treat $\times$ female			-0.277	
			(0.214)	
Treat (cont.) $\times$ female				-0.206***
				(0.061)
Observations	419	419	861	861
R^2	0.250	0.255	0.311	0.314
Panel B: Non-government-related				
	Female		Triple-DiD	
	(5)	(6)	(7)	(8)
Dependent variable: log wage				
Treat	-0.146			
	(0.100)			
Treat (cont.)		-0.049		
		(0.040)		
Treat $\times$ female			-0.195	
			(0.122)	
Treat (cont.) $\times$ female				-0.068
				(0.042)
Observations	1048	1048	2125	2125
R^2	0.223	0.223	0.267	0.267

Notes: The table reports coefficients of interest; all specifications control for years of schooling, age, and age squared. The full regression table is reported in Appendix A.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: Labor market transition matrix (two-year horizon), employed individuals

	Number employed	Unemployed	Employed		Out of labor force
			same employer	changed jobs	
2010–2012	2,367	1.48%	74.31%	13.48%	10.73%
2012–2014	3,327	1.41%	62.79%	23.47%	12.32%
2014–2016	2,925	1.26%	66.56%	20.24%	11.93%
Average	2,873	1.39%	67.89%	19.06%	11.66%

Notes: The data are from China Family Panel Survey. Sample is restricted to all persons older than 16 years old, living in urban areas, appearing in two consecutive waves, and with valid non-farm employment records in both waves.

2.5.3 Which Margin? Worse New Jobs or Stalled Promotions?

Because employment on the extensive margin does not change after the OCP relaxation, the wage decline must operate along the intensive margin. Two natural channels are: (i) new entrants accepting lower-quality jobs (“worse new jobs”), and (ii) slower within-job advancement (“stalled promotion”). I test the promotion channel using CFPS questions on supervisory responsibility—*Has subordinates?* and, if yes, *How many?*. The estimation specification is analogous to Equation (2):

$$\begin{aligned}
Prob.(\text{has subordinates})_{it} &= \beta_0 + \beta_4 \text{Treat}_{pt} \times \text{female}_i \\
&\quad + \mu_p \times \gamma_t + \gamma_t \times \text{female}_i + \mu_p \times \text{female}_i + \varepsilon_{it} \\
\log(\text{num of subordinates})_{it} &= \beta_0 + \beta_4 \text{Treat}_{pt} \times \text{female}_i \\
&\quad + \mu_p \times \gamma_t + \gamma_t \times \text{female}_i + \mu_p \times \text{female}_i + \varepsilon_{it}
\end{aligned}$$

Table 11 shows no evidence of a promotion channel: neither the probability of supervising others nor the (log) number of direct reports changes for young women relative to men. Given the limited sample for counts, these results are suggestive but consistent with the “worse new jobs” interpretation rather than stalled promotions.

Age heterogeneity provides additional evidence on the margin of adjustment. Younger women are more likely to be labor-market entrants or to be forming new employment relationships, whereas women aged 29–35 are more likely to be observed in established jobs. I therefore estimate Equation (2) separately by age group, using age 28 as the cutoff because

Table 11: Effect of OCP Relaxation on Female Workers' Promotions

	Has subordinates		Log subordinates	
	(1)	(2)	(3)	(4)
Female x Treat	0.163 (0.264)		-0.017 (0.400)	
Female x Treat (cont.)		0.096 (0.142)		-0.058 (0.152)
Sample mean	0.20	0.20	1.70	1.70
Mean subordinates			11.92	11.92
R^2			0.208	0.208
Observations	2,003	2,003	514	514

Notes: Standard errors in parentheses. *has_subordinates* is an indicator for holding a supervisory position. *ln_subordinates_num* is the log number of direct reports. *Treat* denotes exposure to the OCP relaxation; *Treat (cont.)* is the continuous exposure measure.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

mean age at childbearing in China was approximately 28 in 2012.⁵

Table 12 shows that the wage decline is concentrated among women aged 18–28, for whom the estimated effect is approximately 21.5%. In contrast, the estimated effect for women aged 29–35 is statistically insignificant. Combined with the absence of changes in supervisory responsibilities, this age pattern is more consistent with lower wage offers or lower-quality job placement among younger workers.

3 Theoretical Model

The empirical results show that young women's wages declined sharply after the relaxation of the One-Child Policy, despite little short-run change in realized fertility or employment. This pattern points to an expectation-based channel: employers revised their beliefs about women's future family responsibilities and priced the implied productivity path into current wage offers.

This section develops a tractable equilibrium framework for that channel. The model

⁵United Nations, Department of Economic and Social Affairs, World Fertility Data (2019).

Table 12: Effect of OCP Relaxation on Female Wages by Age Group

	Panel A: Binary Treatment			Panel B: Cont. Treatment		
	All ages	Age 29–35	Age 18–28	All ages	Age 29–35	Age 18–28
Dependent variable: log wage						
Treat $\times$ female	-0.148*	0.052	-0.363**			
	(0.078)	(0.108)	(0.162)			
Treat (cont.) $\times$ female				-0.075**	0.008	-0.162*
				(0.030)	(0.032)	(0.084)
Observations	2986	1503	1483	2986	1503	1483
R^2	0.239	0.278	0.232	0.239	0.278	0.232

Notes: Standard errors in parentheses. Columns split the main young-worker sample into the full age range, workers ages 29–35, and workers ages 18–28. The binary treatment is the post-OCP relaxation indicator interacted with above-median pre-policy fine rates; the continuous treatment uses the province-level OCP fine rate interacted with the post period. The table reports coefficients of interest; all specifications control for years of schooling, age, and age squared, and include province-by-year, year-by-gender, and province-by-gender fixed effects. The full regression table is reported in Appendix A.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

links household time allocation to wage posting in a frictional labor market with long-term employment relationships. Its purpose is to show how expected fertility can affect wages even when current employment, formal working hours, and initial human capital are held fixed.

The model has two blocks. The household block maps fertility into household work, free time, effective effort, and human-capital accumulation. The labor-market block embeds these gender- and fertility-specific accumulation rates into a wage-posting model with on-the-job search. Firms post fixed wages, workers move only when sufficiently attractive offers arrive, and wages reflect the expected productivity path over the match.

I characterize the firm’s profit function, the equilibrium stock of workers at each posted wage, prove continuity of the offer distribution, and state the equilibrium conditions. The key implication is that unequal household responsibilities can generate gender wage differences through expected productivity growth, even among continuously employed full-time workers.

3.1 Household Collective Decision Model

Household problem. Consider a household with $n \in \{0, 1, 2, \dots\}$ children.⁶

⁶I take n as exogenous in this section in order to focus on the paper’s main mechanism: how socially shared fertility expectations affect household time allocation, human-capital accumulation, and average wage

The household consists of two spouses, a man (M) and a woman (F). Each spouse has a unit time endowment, which is allocated among market work $t_w^j(n)$, household work $t_h^j(n)$, and free time $f^j(n)$, for $j \in \{M, F\}$. The household solves

$$\begin{aligned} \max_{\{t_h^j(n), t_w^j(n)\}_{j \in \{F, M\}}} \quad & \underbrace{\mathbf{1}\{t_w^M(n) \geq \underline{t}\}w^M + \mathbf{1}\{t_w^F(n) \geq \underline{t}\}w^F}_{\text{wage income}} + \underbrace{\phi_L \log(f^M(n) + f^F(n))}_{\text{leisure}} \\ \text{s.t.} \quad & f^j(n) + t_w^j(n) + t_h^j(n) = 1, \quad j \in \{F, M\}, \\ & \gamma^F t_h^F(n) + \gamma^M t_h^M(n) \geq g(n), \quad \gamma^F > \gamma^M. \end{aligned}$$

Here, w^j denotes the full-time wage of spouse j . Employment is modeled as a discrete outcome: a spouse receives the full wage w^j if and only if market work reaches the full-time threshold $\underline{t}$. This threshold can be interpreted as the time requirement for a full-time job, such as an eight-hour workday. Once a spouse works enough hours to be counted as full-time, she or he receives the full wage. Working longer hours, such as staying late or working overtime, does not increase current earnings in the model. This captures the common situation in formal salaried jobs where pay is tied to the employment contract rather than to the exact number of hours worked.

The household also values the total free time of the two spouses, $f^M(n) + f^F(n)$. The parameter $\phi_L > 0$ governs the value of free time relative to wage income. The analysis focuses on parameter values under which full-time employment is attractive in the absence of childcare demands, so that the household does not choose non-employment simply to enjoy more free time. Raising n children requires at least $g(n)$ effective units of household input, where $g(0) = 0$ and $g(n)$ is increasing in n . The parameter γ^j measures the effective contribution of spouse j 's household time to childcare.

Assumption 3.1 (Gender specialization in childcare). The female spouse's household time carries a higher effective weight in childcare: $\gamma^F > \gamma^M$.

setting. This abstraction is motivated by the institutional and social environment in urban China during the sample period. Under the One-Child Policy, urban married couples were typically constrained to have at most one child, while social norms strongly favored childbearing after marriage. Consistent with this, in the CFPS 2010 and 2012 data, 93.2% of urban married or cohabiting individuals ages 32–35 have at least one child, and among those with children, 74.3% have exactly one child. Thus, fertility status in this setting is highly concentrated around one child, which motivates treating n as a predetermined state in the household time-allocation problem.

Assumption 3.1 is a reduced-form way to capture gender-specialized household roles and social norms under which mothers are expected to take greater responsibility for childcare. It is not meant to represent a biological difference in childcare ability. Rather, it summarizes the fact that, in the data and in the relevant social environment, women's time is more closely tied to childcare and household production.

The model has a simple implication for time allocation. Given the step-function wage schedule and the strict value of free time, any employed spouse optimally chooses the minimum market time required for full-time status, $t_w^j(n) = \underline{t}_w$. Hours below $\underline{t}_w$ generate no earnings, while hours above $\underline{t}_w$ do not raise current earnings and only reduce time available for household production or free time. Thus, market attachment is discrete, whereas free time varies continuously with household responsibilities.

With no children, $g(0) = 0$, the childcare constraint is slack. When both spouses work full time, they enjoy the same free time,

$$f^M(0) = f^F(0) = 1 - \underline{t}.$$

Thus, absent children, the household-production channel does not generate gender differences in effort or human-capital accumulation.

For $n > 0$, childcare creates a demand for household time. Conditional on both spouses remaining full-time employed, wage income is fixed at $w^M + w^F$, so the household allocates household work to maximize total free time. Since $\gamma^F > \gamma^M$, household work is allocated first toward the female spouse, subject to her time constraint. Hence, whenever the childcare constraint binds and both spouses remain employed,

$$t_h^F(n) > t_h^M(n) \quad \text{and} \quad f^F(n) < f^M(n).$$

For larger $g(n)$, the household may move to a corner allocation in which one spouse exits full-time employment. The same force makes female non-employment more likely unless the female wage advantage is sufficiently large.

3.2 From Household Time Allocation to Labor-Market Performance

The household block links fertility to labor-market outcomes through workers' available free time. For employed workers, formal market hours are fixed at $t_w^j(n) = \underline{t}$. Hence, differences in career progression among employed workers come from effective work effort rather than hours on the job. Free time affects effort because sleep, rest, and recovery shape workers' energy and productivity during fixed working hours. It also allows for career-related investments outside formal work hours, such as self-study, professional reading, business travel, networking, and client-maintenance activities.

For a worker of gender j with n children, effective effort is given by

$$\varepsilon^j(n) = \frac{1}{1 + \left(\frac{1 - f^j(n) - t_w^j(n)}{\psi_1} \right)^{\psi_2}} = \frac{1}{1 + \left(\frac{t_h^j(n)}{\psi_1} \right)^{\psi_2}}, \quad \psi_1, \psi_2 > 0,$$

where the second equality uses the time constraint. Thus, conditional on full-time employment, greater household work lowers effort, while greater free time raises effort.

Human capital accumulates through learning by doing. I assume that the accumulation rate is linear in effective effort:

$$\alpha^j(n) = \rho_0 + \rho_1 \varepsilon^j(n), \quad \rho_0 > 0, \quad \rho_1 > 0.$$

Thus, fertility can affect future productivity even when current employment status and recorded market hours are unchanged.

Combining the household allocation with the effort and accumulation mappings yields the channel that the labor-market block prices below. When childcare needs increase, household work rises and free time falls. Under Assumption 3.1, women bear a larger share of this adjustment, so the decline in effort and human-capital accumulation is steeper for women.

Proposition 3.2 (Endogenous gender gap in accumulation). *Suppose both spouses remain full-time employed at $t_w^j(n) = \underline{t}$. Under Assumption 3.1, whenever the childcare constraint binds and the household allocation implies $f^F(n) < f^M(n)$,*

$$f^F(n) < f^M(n) \quad \Rightarrow \quad \varepsilon^F(n) < \varepsilon^M(n) \quad \Rightarrow \quad \alpha^F(n) < \alpha^M(n).$$

This implication is consistent with the logic of “greedy jobs,” where continuous availability and off-the-clock investments are rewarded while flexibility is penalized (Goldin, 2014). In the model, free time provides the slack for these effort and investment margins. Because family responsibilities compress women’s free time more than men’s, expected fertility translates into lower expected accumulation rates for women. The labor-market block embeds these gender-specific accumulation rates into wage posting.

Human capital law of motion and output. Given the accumulation rate $\alpha^j(n)$, a worker’s match-specific human capital evolves with tenure τ according to

$$z^j(\tau, n) = z_0 \exp(\alpha^j(n)\tau),$$

where $z_0 > 0$ is the initial human-capital level and $\tau \geq 0$ is job tenure. Human capital is match specific: upon separation, tenure resets to zero and human capital returns to z_0 .

Production only uses effective human capital. For a worker of gender j with tenure τ and n children, output is

$$y^j(\tau, n) = \varepsilon^j(n) z^j(\tau, n). \tag{3}$$

Thus effort affects productivity in two ways. It directly scales the effective use of accumulated human capital, and it indirectly determines the rate at which human capital grows over the match through $\alpha^j(n)$.

3.3 Labor Market Environment

Time is continuous, and I focus on stationary equilibria. The economy consists of two segmented labor markets indexed by gender $j \in \{F, M\}$. In each submarket, there is a continuum of workers of measure m^j and a unit measure of firms. Gender-specific household time allocation generates gender-specific effort and accumulation rates, $\varepsilon^j(n)$ and $\alpha^j(n)$, which may lead to different wage-posting equilibria across submarkets.

Each worker is either unemployed or employed. Unemployed workers receive flow value b^j , while employed workers earn a posted wage w . Job offers arrive according to independent Poisson processes. Unemployed workers receive offers at rate λ_0^j , and employed workers receive

outside offers at rate λ_1^j . Offers are take-it-or-leave-it contracts drawn from a wage-offer distribution $F^j(w)$. The parameters b^j , λ_0^j , and λ_1^j may differ by gender.

Workers follow a reservation rule. Let R^j denote the reservation wage in submarket j : unemployed workers accept offers $w \geq R^j$, and employed workers accept only strictly higher offers. No firm posts below R^j in equilibrium; I therefore take R^j as the lower bound of the offer support and $F^j(R^j) = 0$.

Steady-state flow balance requires inflows into employment to equal exogenous separations from employment:

$$\lambda_0^j [1 - F^j(R^j)] u^j = \delta^j (m^j - u^j), \quad (4)$$

where u^j is the mass of unemployed workers and δ^j is the exogenous separation rate. This condition pins down the size of the unemployment pool for a given offer-arrival environment.

Firms post wages and commit to them. Labor is the only input, and there is no on-the-spot renegotiation or counteroffer after an outside offer arrives.⁷ A worker employed at wage w separates either exogenously at rate δ^j or endogenously after receiving an offer above w . The separation hazard is therefore

$$h^j(w) = \delta^j + \lambda_1^j [1 - F^j(w)].$$

The survival probability in the match up to tenure τ is

$$S^j(\tau | w) = \exp[-h^j(w)\tau].$$

Firm's expected value per hire. A firm that posts wage w commits to that wage for the duration of the match. The wage is fixed, but output changes with tenure because human capital accumulates on the job, as shown in Equation (3). For a worker of gender j and fertility state n , the firm's expected value from a new hire at wage w is

$$\mathbb{E}[\pi^j(w, n)] = \int_0^\infty [\varepsilon^j(n) z_0 \exp(\alpha^j(n)\tau) - w] S^j(\tau | w) d\tau.$$

⁷This “no counteroffer” assumption is standard in on-the-job search models (e.g., [Burdett and Mortensen, 1998](#)) and is also consistent with environments featuring rigid pay scales and limited within-firm bargaining, such as civil-service or state-sector jobs.

This expression discounts future match payoffs by the probability that the worker remains in the match until tenure τ .

Fertility state is not contractible at the wage-posting stage. Let p_n^j denote the firm's predictive probability that a worker in submarket j is in fertility state n over the match horizon. The expected value per hire is

$$\begin{aligned}\mathbb{E}[\pi^j(w)] &= \int_0^\infty \left[\sum_{n=0}^{\bar{n}} \varepsilon^j(n) z_0 \exp(\alpha^j(n)\tau) p_n^j - w \right] S^j(\tau | w) d\tau \\ &= \sum_{n=0}^{\bar{n}} \frac{z_0 \varepsilon^j(n) p_n^j}{h^j(w) - \alpha^j(n)} - \frac{w}{h^j(w)}.\end{aligned}\tag{5}$$

This value is finite whenever $\alpha^j(n) < h^j(w)$ for all n . A sufficient condition is imposed below:

Assumption 3.3 (Finite match value). For every gender j and fertility state n ,

$$\alpha^j(n) < \delta^j.$$

Assumption 3.3 ensures that the expected value of a match is finite uniformly over the wage support, since $h^j(w) \geq \delta^j$ for all posted wages.

Stationary flow profits. The per-hire value in Equation (5) is not yet the firm's payoff from posting wage w . A posted wage affects both the value of each match and the number of matches the firm operates in equilibrium. Higher wages reduce the separation hazard and therefore increase the stock of workers retained by the firm. To aggregate from one hire to firm-level profits, let $q^j(w)$ denote the per-firm inflow rate of new hires at wage w , and let $l^j(w)$ denote the steady-state stock of workers employed by a firm posting w .

A cohort hired at rate $q^j(w)$ survives with probability $S^j(\tau | w)$, so the expected number of workers from that cohort still employed after tenure τ is $q^j(w)S^j(\tau | w)$. In a stationary equilibrium, the stock of workers employed at wage w equals the integral of surviving cohorts:

$$l^j(w) = \int_0^\infty q^j(w) S^j(\tau | w) d\tau = \frac{q^j(w)}{h^j(w)}.$$

This relationship allows the firm's long-horizon cohort payoff to be written as a stationary

flow-profit object. Since $q^j(w) = h^j(w)l^j(w)$, expected flow profits from posting wage w are

$$\begin{aligned}\mathbb{E}[\Pi^j(w)] &= q^j(w)\mathbb{E}[\pi^j(w)] \\ &= \left(\sum_{n=0}^{\bar{n}} \frac{z_0 \varepsilon^j(n) p_n^j h^j(w)}{h^j(w) - \alpha^j(n)} - w \right) l^j(w).\end{aligned}\tag{6}$$

Equilibrium employment stock. It remains to characterize $l^j(w)$. Let $G^j(w, t)$ denote the cross-sectional cdf of accepted wages among employed workers. For $w \geq R^j$, its law of motion is

$$\frac{\partial G^j(w, t)}{\partial t} = \lambda_0^j [F^j(w) - F^j(R^j)] u^j(t) - h^j(w) G^j(w, t) (m^j - u^j(t)).$$

In a stationary equilibrium, using Equation (4),

$$G^j(w) = \frac{\delta^j [F^j(w) - F^j(R^j)]}{h^j(w) [1 - F^j(R^j)]}.$$

The employment stock at a firm posting wage w is obtained by comparing the mass of employed workers earning wages near w with the mass of firms posting wages near w :

$$\begin{aligned}l^j(w) &= \lim_{\epsilon \rightarrow 0} \frac{G^j(w) - G^j(w - \epsilon)}{F^j(w) - F^j(w - \epsilon)} (m^j - u^j) \\ &= \lim_{\epsilon \rightarrow 0} \frac{\delta^j + \lambda_1^j [1 - F^j(R^j)]}{h^j(w) h^j(w - \epsilon)} \lambda_0^j u^j,\end{aligned}$$

where the intermediate algebra is reported in Appendix B.1. Once the no-mass-point property below is established, and using $F^j(R^j) = 0$,

$$l^j(w) = \frac{(\delta^j + \lambda_1^j) \lambda_0^j u^j}{[h^j(w)]^2} = \frac{(\delta^j + \lambda_1^j) \lambda_0^j}{[h^j(w)]^2} \cdot \frac{\delta^j m^j}{\delta^j + \lambda_0^j}.$$

No mass points. The equilibrium wage-offer distribution has no mass points in the interior of its support. If F^j had an atom at an interior wage w , a firm could deviate to $w + \epsilon$. This deviation would discretely lower the separation hazard by placing the firm just above the atom, thereby increasing retention and the employment stock. The wage cost rises only by the infinitesimal amount ϵ , so the deviation is profitable. Hence:

Lemma 3.4 (No mass points in F^j). *Fix $j \in \{F, M\}$. In any stationary wage-posting*

equilibrium, the offer cdf F^j is continuous on $(R^j, \bar{w}^j)$.

The full proof of Lemma 3.4 is provided in Appendix B.2.

Equilibrium firm profit. Substituting $l^j(w)$ into Equation (6) yields

$$\mathbb{E}[\Pi^j(w)] = \frac{(\delta^j + \lambda_1^j)\lambda_0^j u^j}{[h^j(w)]^2} \left(\sum_{n=0}^{\bar{n}} \frac{z_0 \varepsilon^j(n) p_n^j h^j(w)}{h^j(w) - \alpha^j(n)} - w \right). \quad (7)$$

Equation (7) contains the standard Burdett–Mortensen retention margin and the new expected-productivity channel. The standard component is $l^j(w) \propto [h^j(w)]^{-2}$: higher wages reduce separations to better offers and allow firms to retain a larger workforce. The new component is

$$\sum_{n=0}^{\bar{n}} \frac{z_0 \varepsilon^j(n) p_n^j h^j(w)}{h^j(w) - \alpha^j(n)}.$$

This term is absent from the canonical model. It appears because output grows over the match at fertility-state-specific rate $\alpha^j(n)$, and firms price the expected productivity path using beliefs p_n^j . If $\alpha^j(n) = 0$ and $\varepsilon^j(n) = 1$ for all n , the expression collapses to the standard constant-productivity Burdett–Mortensen case.

Because each firm is infinitesimal, it takes the aggregate offer distribution F^j as given when choosing its posted wage. The equilibrium offer distribution must then be generated by firms’ best responses. Hence, in any mixed-strategy wage-posting equilibrium, all wages posted with positive probability yield the same expected profit; otherwise firms would shift mass toward more profitable wages. Wages outside the support yield weakly lower profits.

3.4 Workers’ Problem

The worker side follows the standard on-the-job search structure. Workers discount future utility at rate $r > 0$. Let V_0^j denote the value of unemployment in gender submarket j , and let $V_1^j(w)$ denote the value of employment at wage w . An unemployed worker receives flow value b^j and receives job offers at rate λ_0^j . Hence,

$$rV_0^j = b^j + \lambda_0^j \int_{R^j}^{\bar{w}^j} [V_1^j(x) - V_0^j] dF^j(x).$$

An employed worker earns wage w , receives outside offers at rate λ_1^j , and separates exogenously at rate δ^j . The value of employment satisfies

$$rV_1^j(w) = w + \lambda_1^j \int_w^{\bar{w}^j} [V_1^j(x) - V_1^j(w)] dF^j(x) + \delta^j [V_0^j - V_1^j(w)], \quad w \in [R^j, \bar{w}^j].$$

Since $V_1^j(w)$ is increasing in w , the optimal acceptance rule is characterized by a reservation wage R^j satisfying

$$V_1^j(R^j) = V_0^j.$$

Unemployed workers accept offers $w \geq R^j$, and employed workers accept only offers strictly above their current wage.

Evaluating the employed-worker Bellman equation at $w = R^j$, using $V_1^j(R^j) = V_0^j$, and combining it with the unemployment Bellman equation gives the reservation-wage condition

$$\begin{aligned} R^j - b^j &= (\lambda_0^j - \lambda_1^j) \int_{R^j}^{\bar{w}^j} [V_1^j(x) - V_0^j] dF^j(x) \\ &= (\lambda_0^j - \lambda_1^j) \int_{R^j}^{\bar{w}^j} \frac{1 - F^j(x)}{r + \delta^j + \lambda_1^j [1 - F^j(x)]} dx. \end{aligned} \tag{8}$$

This condition determines the reservation wage given the offer distribution.

3.5 Equilibrium

I now collect the worker problem, firm best responses, and steady-state flow conditions into the equilibrium definition.

Definition 3.5 (Stationary wage-posting equilibrium). Fix gender $j \in \{F, M\}$ and parameters

$$(m^j, r, b^j, \lambda_0^j, \lambda_1^j, \delta^j, z_0, \{\varepsilon^j(n), \alpha^j(n), p_n^j\}_{n=0}^{\bar{n}}).$$

A stationary wage-posting equilibrium in submarket j is a tuple

$$(F^j, G^j, u^j, R^j, \bar{w}^j),$$

where F^j is the cdf of posted wage offers, G^j is the cross-sectional cdf of accepted wages among employed workers, u^j is the mass of unemployed workers, R^j is the reservation wage,

and $\bar{w}^j$ is the upper bound of the offer support, such that:

- (i) **Worker optimality.** Workers follow the reservation and job-ladder rules described in Section 3.4: unemployed workers accept offers $w \geq R^j$, employed workers accept only strictly higher offers, and R^j satisfies Equation (8).
- (ii) **Firm optimality.** Taking F^j and the induced hazard

$$h^j(w) = \delta^j + \lambda_1^j [1 - F^j(w)]$$

as given, firms choose posted wages to maximize expected flow profits

$$\mathbb{E}[\Pi^j(w; F^j)] = \frac{(\delta^j + \lambda_1^j)\lambda_0^j u^j}{[h^j(w)]^2} \left(\sum_{n=0}^{\bar{n}} \frac{z_0 \varepsilon^j(n) p_n^j h^j(w)}{h^j(w) - \alpha^j(n)} - w \right).$$

There exists an equilibrium profit level $\bar{\pi}^j$ such that

$$\mathbb{E}[\Pi^j(w; F^j)] = \bar{\pi}^j \quad \text{for all } w \in \text{supp}(F^j),$$

$$\mathbb{E}[\Pi^j(w; F^j)] \leq \bar{\pi}^j \quad \text{for all feasible } w \in [R^j, \bar{w}^j].$$

- (iii) **Support and regularity.** The offer distribution F^j is supported on $[R^j, \bar{w}^j]$, with the lower endpoint normalized as $F^j(R^j) = 0$ and $F^j(\bar{w}^j) = 1$. The no-mass-point property in Lemma 3.4 holds on the interior of the support.
- (iv) **Flow balance and cross-sectional consistency.** Steady-state employment flows satisfy

$$\lambda_0^j [1 - F^j(R^j)] u^j = \delta^j (m^j - u^j),$$

and the accepted-wage distribution satisfies

$$G^j(w) = \frac{\delta^j [F^j(w) - F^j(R^j)]}{h^j(w) [1 - F^j(R^j)]}, \quad w \in [R^j, \bar{w}^j].$$

- (v) **Stationarity.** All equilibrium objects are time invariant and mutually consistent with the offer distribution F^j , the accepted-wage distribution G^j , the unemployment mass u^j , and the hazard schedule $h^j(w)$.

4 Solution Method and Estimation Results

This section brings the model to the data and quantifies the role of expected productivity in labor markets with contracted labor. I recast the equilibrium conditions in a form suitable for computation, describe the algorithm that recovers the offer distribution from parameters, and estimate these parameters by matching the model-implied wage CDFs to the CFPS distributions (2012 baseline and 2014 post-reform). I show the baseline fit, re-estimate the model after the policy relaxation, and conduct counterfactuals that (i) equalize family-work time, and (ii) update only fertility expectations to quantify how expectation-driven pricing accounts for both the baseline gap and its post-reform widening.

4.1 Numerical Solution

This subsection rewrites the equilibrium conditions from Section 3 in a form that can be solved numerically. I fix a gender submarket j and suppress the gender superscript. For a given parameter vector

$$\theta = (r, b, \delta, \lambda_0, \lambda_1, z_0, \{\varepsilon(n), \alpha(n), p_n\}_{n=0}^{\bar{n}}),$$

the endogenous objects are the reservation wage R , the upper support point $\bar{w}$, and the wage-offer distribution F . Under Assumption 3.3, firm values are finite over the full support.

The first step is to determine the support of the offer distribution. For any candidate reservation wage R , the lower endpoint satisfies $F(R) = 0$, so $h(R) = \delta + \lambda_1$. The upper endpoint satisfies $F(\bar{w}) = 1$, so $h(\bar{w}) = \delta$. Evaluating the firm profit expression from Section 3 at these two endpoints and imposing profit equalization, $\Pi(R) = \Pi(\bar{w})$, gives

$$\bar{w}(R) = \sum_{n=0}^{\bar{n}} \frac{z_0 \varepsilon(n) p_n \delta}{\delta - \alpha(n)} - \frac{\delta^2}{(\delta + \lambda_1)^2} \left[\sum_{n=0}^{\bar{n}} \frac{z_0 \varepsilon(n) p_n (\delta + \lambda_1)}{\delta + \lambda_1 - \alpha(n)} - R \right]. \quad (9)$$

Given R and $\bar{w}(R)$, the interior of the offer distribution is recovered from the same equal-profit condition. For each $w \in (R, \bar{w})$, solve for $h(w) \in (\delta, \delta + \lambda_1)$ from

$$\frac{1}{[h(w)]^2} \left[\sum_{n=0}^{\bar{n}} \frac{z_0 \varepsilon(n) p_n h(w)}{h(w) - \alpha(n)} - w \right] = \frac{1}{(\delta + \lambda_1)^2} \left[\sum_{n=0}^{\bar{n}} \frac{z_0 \varepsilon(n) p_n (\delta + \lambda_1)}{\delta + \lambda_1 - \alpha(n)} - R \right]. \quad (10)$$

The offer cdf then follows from the hazard definition:

$$F(w) = 1 - \frac{h(w) - \delta}{\lambda_1}.$$

The worker side closes the model. Given F , the reservation wage must satisfy

$$R - b = (\lambda_0 - \lambda_1) \int_R^{\bar{w}} \frac{1 - F(x)}{r + \delta + \lambda_1[1 - F(x)]} dx. \quad (11)$$

The numerical solution therefore reduces the equilibrium computation to a one-dimensional root-finding problem in R .

Algorithm. For a given parameter vector θ , solve for the equilibrium objects as follows. For each trial value of R , compute the implied upper support point $\bar{w}(R)$ using Equation (9). On a grid of wages $\{w_k\}_{k=1}^K \subset [R, \bar{w}(R)]$, solve Equation (10) for $h(w_k)$ at each interior node and recover $F(w_k)$ from the hazard definition. I impose the endpoint values $F(R) = 0$ and $F(\bar{w}) = 1$.

A trial value of R is admissible only if the recovered F is a valid cdf on $[R, \bar{w}(R)]$. For admissible trial values, interpolation over $\{w_k, F(w_k)\}_{k=1}^K$ yields a candidate offer distribution $F(\cdot; R, \theta)$.

Using this candidate F , evaluate the reservation residual

$$\Phi(R; \theta) \equiv (R - b) - (\lambda_0 - \lambda_1) \int_R^{\bar{w}(R)} \frac{1 - F(x; R, \theta)}{r + \delta + \lambda_1[1 - F(x; R, \theta)]} dx.$$

The equilibrium reservation wage R^* solves

$$\Phi(R^*; \theta) = 0.$$

The associated $\bar{w}(R^*)$ and $F(\cdot; R^*, \theta)$ are the equilibrium upper support point and offer distribution.

Finally, the model-implied cross-sectional wage distribution among employed workers is obtained from the Burdett–Mortensen stock-flow mapping:

$$G(w) = \frac{\delta F(w)}{\delta + \lambda_1[1 - F(w)]} \iff F(w) = \frac{(\delta + \lambda_1)G(w)}{\delta + \lambda_1 G(w)}. \quad (12)$$

The estimation compares this model-implied $G(\cdot; \theta)$ with the empirical wage CDF.

4.2 Identification Argument

The numerical solution in the previous subsection describes the forward problem: given parameters, solve for the endogenous reservation wage R , offer distribution F , and employed-wage distribution G . This subsection discusses the inverse problem. Given the observed wage distribution, what features of the model parameters are disciplined by the equilibrium restrictions?

I start with an average-productivity version only as an expositional benchmark, not as a separate estimation step. This benchmark shows how the wage distribution disciplines λ_1 together with aggregate productivity growth, in the same spirit as the canonical Burdett–Mortensen model. I then return to the full model, where the aggregate productivity term is replaced by a finite sum over fertility states; conditional on (δ, λ_1) , the same rational-function logic separates the state-specific terms. Finally, the observed unemployment rate, through steady-state flow balance, identifies the unemployed offer-arrival rate λ_0 , and the reservation-wage condition then identifies the unemployment flow value b .

Given δ and a candidate value of λ_1 , the employed-wage distribution $G(w)$ implies the offer distribution and the separation hazard:

$$F(w) = \frac{(\delta + \lambda_1)G(w)}{\delta + \lambda_1 G(w)}, \quad h(w) = \frac{\delta(\delta + \lambda_1)}{\delta + \lambda_1 G(w)}.$$

Collapse fertility heterogeneity into two average objects:

$$\bar{\alpha} \quad \text{and} \quad \bar{\beta} \equiv z_0 \bar{\epsilon}.$$

With these average objects, profit equalization between the lower support point R and any observed wage w implies

$$w = \frac{\bar{\beta}h(w)}{h(w) - \bar{\alpha}} - \frac{[h(w)]^2}{(\delta + \lambda_1)^2} \left[\frac{\bar{\beta}(\delta + \lambda_1)}{\delta + \lambda_1 - \bar{\alpha}} - R \right]. \quad (13)$$

The left-hand side is observed, while the right-hand side is a parametric function of $G(w)$ through $h(w)$. Conditional on δ , the requirement that Equation (13) hold over the wage support identifies $(\lambda_1, \bar{\alpha}, \bar{\beta})$ under nondegenerate support variation. Appendix B.3 provides the proof for the average-productivity case. In the special case $\bar{\alpha} = 0$, Equation (13) collapses

to the canonical constant-productivity wage-posting restriction.

The full model uses the same equality but replaces the single average-productivity term with a sum over fertility states. Once (δ, λ_1) are fixed, $G(w)$ recovers $F(w)$ and $h(w)$. Profit equalization between the lower support point R and a generic wage w gives

$$w = \frac{[h(w)]^2}{(\delta + \lambda_1)^2} R + \sum_{n=0}^{\bar{n}} z_0 \varepsilon(n) p_n \left[\frac{h(w)}{h(w) - \alpha(n)} - \frac{[h(w)]^2}{(\delta + \lambda_1)[\delta + \lambda_1 - \alpha(n)]} \right]. \quad (14)$$

Suppose the fertility-state probabilities p_n are observed from the data. The second term in Equation (14) is a finite sum of rational functions of $h(w)$. Under Assumption 3.3, and when the nonzero $\alpha(n)$'s are distinct, the pole-and-coefficient logic identifies the state-specific accumulation rates $\alpha(n)$ and productivity scales $z_0 \varepsilon(n)$, up to relabeling. Because fertility states are labeled in the data and p_n is attached to those states, the matching is pinned down by state labels. With a normalization such as $\varepsilon(0) = 1$, z_0 is identified from $z_0 \varepsilon(0)$, and the remaining effort levels follow from $\varepsilon(n) = [z_0 \varepsilon(n)]/z_0$. Appendix B.4 provides the formal proof.

The remaining parameters (λ_0, b) are identified from the observed unemployment rate and the reservation-wage condition. Using the observed unemployment rate u/m , steady-state employment flow balance gives

$$\lambda_0 u = \delta(m - u), \quad \text{so} \quad \lambda_0 = \delta \frac{1 - u/m}{u/m}.$$

Given λ_0 , the reservation-wage equation identifies b . Defining

$$C(F; \delta, \lambda_1, r) \equiv \int_R^{\bar{w}} \frac{1 - F(x)}{r + \delta + \lambda_1[1 - F(x)]} dx,$$

the reservation condition implies

$$b = R - (\lambda_0 - \lambda_1)C(F; \delta, \lambda_1, r).$$

Thus the unemployment moment identifies λ_0 , and the reservation condition then identifies b .

Summary. The average-productivity restriction shows how the wage distribution disciplines λ_1 , average accumulation, and average effective productivity in the same spirit as the canonical

wage-posting model. Conditional on the search parameters (δ, λ_1) , the richer fertility-state structure further identifies state-specific $\alpha(n)$ and $z_0\varepsilon(n)$ under distinctness and normalization conditions.

Finally, the unemployment rate and the reservation-wage condition identify λ_0 and b . The estimation below implements this logic by fixing δ , using observed fertility-state probabilities and unemployment moments, and choosing the remaining parameters to match the full observed wage distribution.

4.3 Estimation

In this subsection, I use Chinese wage data from 2012 and 2014 to estimate the parameters governing the equilibrium wage distribution before and after the policy relaxation. The analysis serves three purposes. First, I estimate the pre-policy parameters to characterize the labor markets for women and men and to decompose the residual gender wage gap—after controlling for observables—to assess how much of it can be explained by the lower expected-productivity channel, which is the focus of this paper. Second, I re-estimate the parameters using post-policy data, treating the labor market as a black box, to infer how employers adjusted expectations and offer behavior following the policy change, and thus, to check whether the expectation channel really matters.

I retain the sample used in the difference-in-differences analysis in Section 2 and construct gender-specific wage distributions before and after the OCP relaxation (2012 and 2014). For each gender-year cell, I estimate the parameter vector

$$\theta_t^j = \left(\{\alpha^j(n)\}_n, \lambda_{0,t}^j, \lambda_{1,t}^j, \delta^j, b^j, z_{0,t}, \{\varepsilon^j(n)\}_n \right)$$

by squared minimum distance (L2 norm) between the model-implied employed-wage distribution, $G(\cdot; \theta)$, and the empirical CDF $\hat{G}(\cdot)$. The model assumes that both genders share a common initial human capital level in year t , denoted $z_{0,t}$. Accordingly, I work with residual wages net of year⁸, province, education, and industry fixed effects to align the data with

⁸Controlling for year fixed effects removes average wage differences across years but does not ensure that the normalized initial human capital $z_{0,t}$ is constant over time. In practice, $z_{0,t}$ may still vary nonlinearly across years, which the estimation allows to capture.

the model's normalization of z_0 . The assumption of a gender-common $z_{0,t}$ is motivated by the fact that gender differences in educational attainment have largely disappeared or even reversed in China. Thus, there is no empirical basis to assume that young women graduating from the same or better universities start with a lower level of human capital than their male peers.

Except for the common $z_{0,t}$, all other parameters are gender-specific, including the offer arrival rates while employed ($\lambda_{1,t}^j$) and while unemployed ($\lambda_{0,t}^j$), the unemployment benefits (b^j), the exogenous separation rates (δ^j), the human capital accumulation rates conditional on marriage and fertility status ($\alpha^j(n)$), and the effort levels ($\varepsilon^j(n)$). I assume that the unemployment benefits b^j and the exogenous separation rates remain constant over time, and that the mappings from n to effort, $\varepsilon(n)$, and from effort to human capital accumulation, $\alpha(n)$, are stable across survey years.

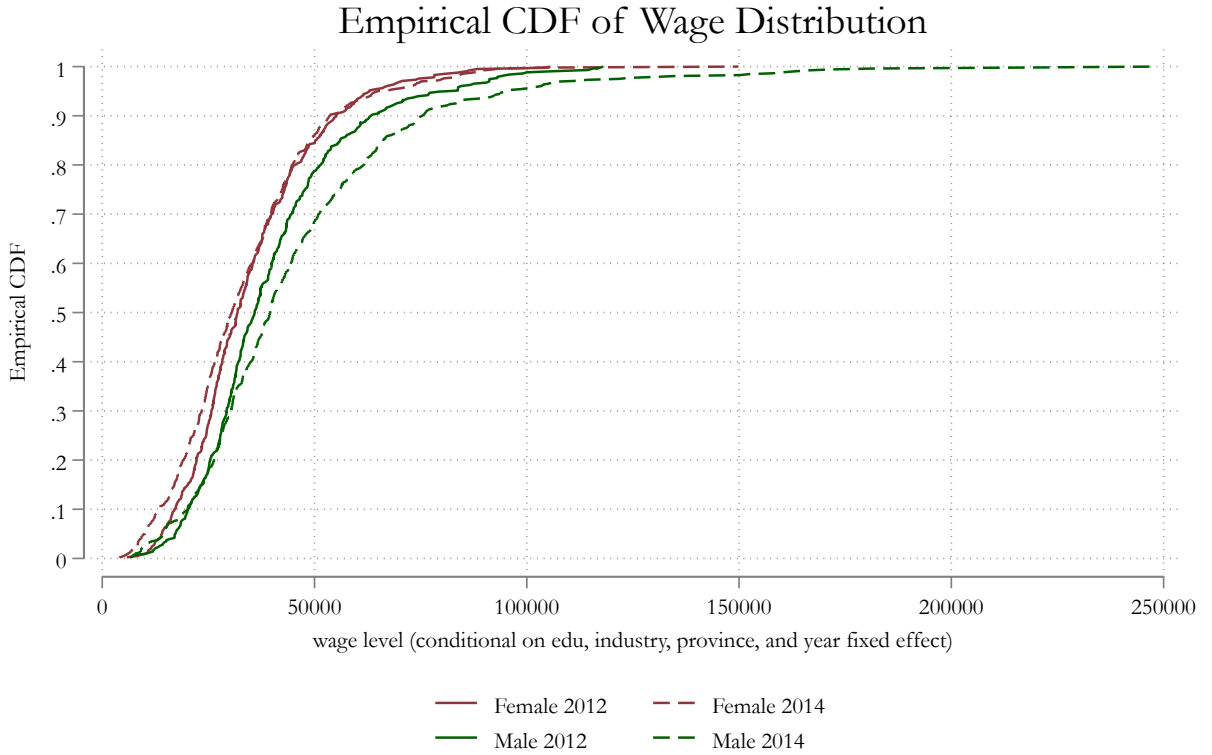


Figure 4: Employed-wage CDFs before and after the OCP relaxation

Figure 4 plots the empirical wage CDFs: solid lines represent the pre-OCP period (2012), and dashed lines represent the post-OCP period (2014). Two patterns stand out. First,

the female CDF shifts only slightly overall, with a modest leftward movement in the lower quantiles. This suggests that new female entrants to the labor market found it harder to obtain high-paying jobs after the policy change and thus accepted lower wages. Second, the male CDF shifts noticeably to the right, particularly in the upper quantiles, indicating that men’s wages rose, especially among high earners. Examining the full distribution, rather than focusing solely on average wages, allows the analysis to extract more information from a limited sample and to identify which parts of the wage distribution drive the observed changes in average earnings identified in the empirical part.

4.3.1 Calibration for Pre-Determined Parameters

Consistent with the identification discussion in Section 4.2, a few parameters are fixed exogenously so that the remaining objects are identified within the model.

Discounting rate. I set the annual discount rate to $r = 0.05$, as is standard in the literature.

Exogenous separation. In a wage-posting environment where firms do not fire workers within matches, “separation” is best interpreted as match termination for reasons outside the firm’s control. In China’s market for highly educated workers—where government-related jobs are prevalent and tenure-like employment security is common—this primarily reflects exits from the labor force or from the estimation window (e.g., aging out of the 18–35 sample), rather than displacement. I therefore set δ equal to the demography-driven “age-out” hazard.⁹

In steady state, the outflow from employment equals the inflow from unemployment: $\delta^j(m^j - u^j) = \lambda_0^j u^j$. Interpreting δ^j as the age-out rate implies that, each year, $\delta^j(m^j - u^j)$ workers leave employment and an equivalent mass of new graduates enters the labor force, keeping m^j stable. Likewise, for the unemployment pool, $\lambda_0^j u^j$ workers are hired and replaced by $\delta^j(m^j - u^j)$ new entrants, so u^j remains stationary. Treating δ^j as an “age-out” hazard introduces a life-cycle element into this otherwise static equilibrium model. New entrants—recent graduates entering the labor market—begin as temporarily unemployed, receive an initial job offer, and remain in that match for the rest of their careers unless they receive and accept a better wage offer, analogous to a promotion.

⁹This “age-out” separation rate should be viewed as a lower bound for the true δ , since it excludes exits due to health, family, or other personal reasons—including cases where women leave the labor market to become full-time homemakers or caregivers.

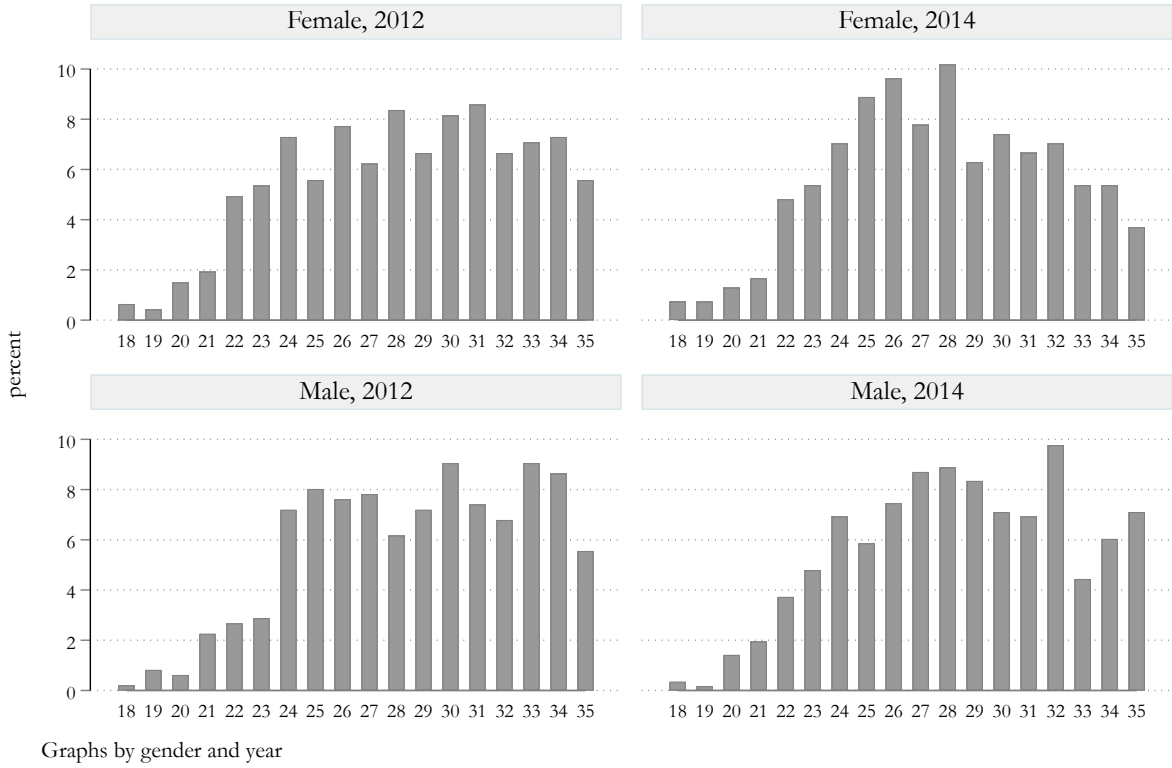


Figure 5: Age Distribution of Women and Men in the Sample (Ages 18–35). *Notes: The sample matches that used in the empirical analysis (Section 2): currently employed urban residents ages 18–35 with at least a high-school education.*

Figure 5 presents the age distribution of employed men and women aged 18–35 in 2012 and 2014. In both years, the distributions are broadly similar across genders, with most workers concentrated between their mid-20s and early 30s. Between 2012 and 2014, the overall shape of the distribution remains stable, though the female sample becomes slightly more concentrated among individuals aged 24 to 28, reflecting the entry of younger cohorts into the labor force.

To calibrate the exogenous separation rate δ , I proxy it with the average share of employed workers at terminal ages (33–35) within the 18–35 window.¹⁰ The separation rate for women is consistently lower than that for men. Because δ determines the upper bound of the human-capital accumulation rate $\alpha(n)$, I hold it constant across years to isolate the change

¹⁰In 2012, the shares for women aged 33, 34, and 35 are 7.08%, 7.30%, and 5.58%, implying $\delta^F_{2012} = 6.65\%$. For men, the corresponding shares are 9.05%, 8.64%, and 5.56%, giving $\delta^M_{2012} = 7.75\%$. In 2014, the female shares are 5.37%, 5.37%, and 3.70% ($\delta^F_{2014} = 4.81\%$), while the male shares are 4.44%, 6.04%, and 7.10% ($\delta^M_{2014} = 5.86\%$).

in $\alpha(n)$. Specifically, I set $\delta^F = 6.65\%$ and $\delta^M = 7.75\%$ ¹¹.

Marriage and fertility-state probabilities. I calibrate the expected marriage and fertility-state probabilities, p_n^j , defined as the probabilities employers assign—at hiring—that a representative worker of gender j will spend time in a given marriage or fertility state n over the expected match horizon. Because age is not explicitly modeled as a state variable, I assume firms do not observe exact age and instead apply a flat prior over the hiring-age window $[a_L, a_H] = [18, 35]$. The observed fractions of individuals in each state within the sample thus govern the calibration of p_n^j . The states are defined as follows: $n = 0$ denotes unmarried and childless; $n = 1$ denotes married without children; and $n \geq 2$ denotes married with $n - 1$ children. ¹²

Using the 2012 CFPS sample restricted to individuals in the labor force, under age 35, and with at least a high-school education, I compute the empirical distribution of these states. Among women, 31% are unmarried and childless ($n = 0$), 18% are married without children ($n = 1$), and 51% have at least one child ($n \geq 2$). ¹³ Only 3.23% of women and 4.53% of men report having two children before the policy relaxation; therefore, I treat all parents as belonging to a single “married with one child” category ($n = 2$). Among men, the corresponding shares are 38% for $n = 0$, 17% for $n = 1$, and 45% for $n = 2$.

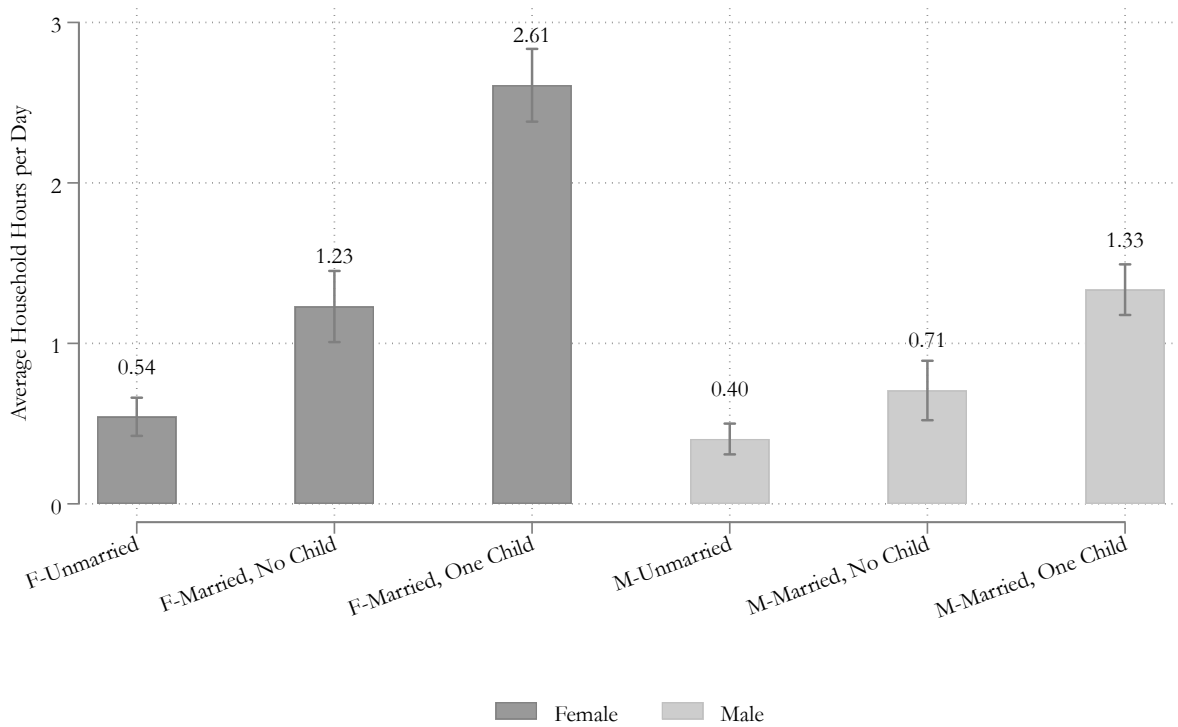
Effort and accumulation mappings. Identification implies that $\alpha^j(n)$ and $\varepsilon^j(n)$ are determined up to a permutation of n . Building on the household block introduced in Section 3.1, I assume both genders share a common mapping from family-work time $t^j(n)$ to work effort and human-capital accumulation rates: $\varepsilon^j(n) = \frac{1}{1+(t^j(n)/\psi_1)^{\psi_2}} \in (0, 1]$, $\psi_1 > 0$, $\psi_2 > 0$, and $\alpha^j(n) = \rho_0 + \rho_1 \varepsilon^j(n)$, $\rho_0 > 0$, $\rho_1 > 0$. Both $\varepsilon^j(n)$ and $\alpha^j(n)$ increase with available free time and thus decrease with family-work time. Under this specification, individuals with no family-work burden ($t^j(n) = 0$) exert full effort, $\varepsilon^j(n_0) = 1$.

Free time is not directly observed, so I proxy it by the complement of family-work time $t^j(n)$ —the sum of daily hours spent on household chores and family care—using data from the CFPS time-use module. The reproduced CFPS weekday averages indicate that, among

¹¹I use the higher estimates because the observed age-out rate represents a lower bound, and a slightly larger δ provides a more conservative calibration

¹²In the data, individuals with children are never observed as unmarried.

¹³Cohabiting and divorced individuals are grouped with the “married” category because they represent a small share of the sample—2.38% for women and 1.3% for men.



CFPS 2010: Urban individuals aged 18–35, waged employment, $\geq$ high school education. Error bars indicate 95% confidence intervals.

Figure 6: Average Household Hours by Gender and Family Status.

unmarried individuals, men spend about 0.41 hours per day on family work, while women spend slightly more, around 0.54 hours. After marriage, men’s average family-work time rises modestly to 0.73 hours per day, whereas women’s time nearly doubles to 1.23 hours even before having children. When a child is present, men devote about 1.32 hours per day to family work, while women’s time increases sharply to 2.60 hours. This stark divergence highlights a key feature of the Chinese social context: women bear the primary responsibility for household chores and childcare. Consequently, their available free time—and thus effective work effort and human-capital accumulation—declines more steeply with family formation compared to men.

Summary of externally calibrated inputs:

$$r = 0.05, \delta^F = 0.0665, \delta^M = 0.0775,$$

$$p_{0,2012}^F = 0.31, p_{1,2012}^F = 0.18, p_{2,2012}^F = 0.51; p_{0,2012}^M = 0.38, p_{1,2012}^M = 0.17, p_{2,2012}^M = 0.45,$$

$$t^F(0) = 0.54, t^F(1) = 1.23, t^F(2) = 2.60; t^M(0) = 0.41, t^M(1) = 0.73, t^M(2) = 1.32.$$

4.4 Estimation for 2012

The estimation algorithm follows a standard minimum-distance approach. Given the pre-determined parameters discussed in the previous subsection, $\{r, \delta^j, \{p_n^j\}_n, \{t^j(n)\}_n\}$, I begin by guessing an initial parameter vector $\theta^{j,0} = (\{\alpha(n)\}_n, \lambda_0, \lambda_1, \delta, b, z_0, \{\varepsilon(n)\}_n)$ for each gender j . For a given guess $\theta^{j,0}$, I solve the model to obtain the implied wage-offer distribution $F^j(w; \theta^{j,0})$, convert it to the employed-wage distribution via

$$G^j(w; \theta^{j,0}) = \frac{\delta^j F^j(w; \theta^{j,0})}{\delta^j + \lambda_1^j (1 - F^j(w; \theta^{j,0}))},$$

and then compare the model-implied $G^j(w; \theta^{j,0})$ with the empirical wage CDF $\widehat{G}^j(w)$ using the *Anderson–Darling (AD) distance*. This procedure is iterated until the sum of AD distance for women and men is minimized.

The AD distance measures the weighted discrepancy between the model-implied and empirical CDFs, placing more weight on the tails where wage data contain the most information about mobility, promotions, and inequality. This feature is desirable in the present setting, as the data exhibit that most changes are in the tails after the policy shifts. Formally, for an empirical CDF $\widehat{G}^j(w)$ and a model-implied CDF $G^j(w; \theta)$ on the support $[R^j, \overline{w}^j]$, the AD distance for gender j is

$$\widehat{AD}^j(\theta) = \int_{R^j}^{\overline{w}^j} \frac{[G^j(w; \theta) - \widehat{G}^j(w)]^2}{\widehat{G}^j(w) [1 - \widehat{G}^j(w)]} d\widehat{G}^j(w).$$

In practice, this integral is evaluated numerically over a fine grid of wage quantiles, with the weighting term $[\widehat{G}^j(w) (1 - \widehat{G}^j(w))]^{-1}$ automatically emphasizing the tails. The estimator $\widehat{\theta}$

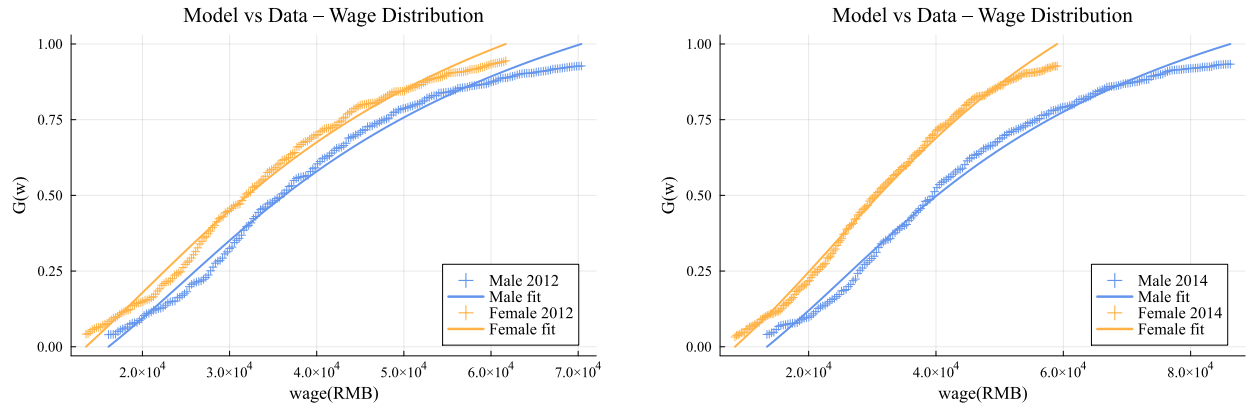
best reproduces the observed wage distributions for men and women, formally

$$\hat{\theta} = \arg \min_{\theta} \widehat{AD}(\theta) = \sum_{j \in \{F, M\}} \widehat{AD}^j(\theta)$$

Under certain M -estimation regularity conditions (Newey and McFadden (1994)), the AD estimator is consistent and asymptotically normal:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \Omega(\theta_0)).$$

where n is the sample size. Standard errors $se(\hat{\theta})$ can be computed from sample covariance matrix $\hat{\Omega}$, so does the 95% confidence interval $[\hat{\theta}_i - 1.96 se(\hat{\theta}_i), \hat{\theta}_i + 1.96 se(\hat{\theta}_i)]$.



(a) 2012: empirical vs. model CDFs by gender

(b) 2014: empirical vs. model CDFs by gender

Figure 7: Model fit for employed wage distributions

The estimation results for the 2012 gender-specific wage distributions are summarized in Table 13, and the corresponding model fit is presented in Figure 7.

Panel A of Table 13 reports the estimated parameters linking family-work time to work effort (ψ_1, ψ_2) and work effort to human-capital accumulation (ρ_0, ρ_1), as defined by

$$\varepsilon^j(n) = \frac{1}{1 + (t^j(n)/\psi_1)^{\psi_2}} \in (0, 1], \quad \psi_1 > 0, \quad \psi_2 > 0,$$

and

$$\alpha^j(n) = \rho_0 + \rho_1 \varepsilon^j(n), \quad \rho_0 > 0, \quad \rho_1 > 0.$$

The positive estimates $\psi_1 > 0$ and $\psi_2 > 0$ confirm that greater family-work time reduces effective work effort. Likewise, $\rho_1 = 0.03 > 0$ implies that a worker exerting full effort

($\varepsilon^j(n) = 1$) accumulates human capital at a rate roughly 0.03 above the baseline $\rho_0 = 0.02$, corresponding to an annual firm-specific accumulation rate of about 5%.

Substituting the observed family-work time values, unmarried men and women exert nearly full effort. However, once married, the gender gap in work intensity begins to widen: men’s effort remains close to 99.8%, whereas women’s declines slightly to about 98.5%. The most pronounced divergence occurs with the arrival of the first child. Parenthood lowers men’s effective effort modestly—to around 97.9%—but substantially reduces women’s to roughly 62%. Consequently, women with children experience much slower human-capital accumulation than childless women, while the reduction for fathers is relatively small. This pattern mirrors the unequal division of household responsibilities observed in time-use data and underpins the model’s mechanism that links expected family labor burdens to gendered wage differentials.

Table 13: Results for Estimated Parameters in Year 2012

Panel A: Human Capital Accumulation						
<i>Accumulation mapping parameters:</i> $\psi_1 = 2.88$, $\psi_2 = 4.98$, $\rho_0 = 0.02$, $\rho_1 = 0.03$						
$n = 0$: unmarried $n = 1$: married without kids $n = 2$: with one kid						
	$e(0)$	$e(1)$	$e(2)$	$\alpha(0)$	$\alpha(1)$	$\alpha(2)$
Female	0.999	0.985	0.621	0.052	0.051	0.040
Male	0.999	0.998	0.979	0.052	0.052	0.051

Panel B: Labor Market Parameters				
2012	b	λ_0	λ_1	z_0
Female	3,874 (3,849, 3,898)	0.512 (0.495, 0.529)	0.403 (0.374, 0.432)	23,352 (23,312, 23,391)
Male	3,818 (3,795, 3,842)	0.614 (0.556, 0.672)	0.506 (0.441, 0.571)	23,352 (23,312, 23,391)

Notes: Panel A reports effort $e(n)$ and human-capital accumulation rates $\alpha(n)$ for different marriage and fertility status n . Panel B reports estimated labor market parameters: unemployment benefits (b), job-offer arrival rates for unemployed (λ_0) and employed (λ_1) workers, and the common initial productivity level (z_0). All values in parentheses represent 95% confidence intervals. Monetary values are expressed in RMB per year.

Panel B of Table 13 reports the estimated labor-market parameters that characterize China’s market for highly educated workers. The estimated initial human capital level, $z_0 = 23,300$ RMB per year, represents the market price for initial human-capital level. For

comparison, the average nominal wage among employed individuals in the 2012 sample is 30,142 RMB, with a median of 27,000 RMB. Given the estimated accumulation rate of about 5% per year, a worker’s productivity after five years of tenure would rise to roughly 29,700 RMB, and after ten years to about 37,900 RMB. Considering the low mobility typical of China’s high-skill labor market, long tenures of ten years or more are not uncommon, making such cumulative productivity gains plausible.

The estimated unemployment benefit level, $b^j \approx 3,800$ RMB for both men and women, reflects not only formal unemployment insurance but also the imputed monetary value of leisure and home production. In equilibrium, the reservation wage R —the lowest wage an individual is willing to accept—must exceed this value. Consistent with this, the smallest observed wage in the sample is 3,000 RMB, and fewer than 1% of respondents report wages below 3,500 RMB.

Finally, the two job-offer arrival rates, λ_0^j and λ_1^j , capture market mobility while unemployed and employed, respectively. The estimates satisfy $\lambda_0^F < \lambda_0^M$ and $\lambda_1^F < \lambda_1^M$, indicating that women receive fewer job opportunities both when unemployed and when employed. This pattern is consistent with a less dynamic job ladder and more constrained career mobility for women relative to men.

The magnitudes of λ_0^j and λ_1^j are broadly comparable to those found in the search-and-matching literature, though direct comparisons are disputable by sample and model differences. For instance, [Postel-Vinay and Robin \(2002\)](#), using French administrative data for 1996–1998, report annual on-the-job offer-arrival rates (λ_1) between 0.64 and 0.67 across industries—slightly above but of similar order to the estimates obtained here.

To further evaluate whether the estimated offer-arrival rates are consistent with dynamic features of China’s labor market, I use the panel dimension of the CFPS to compute employer-to-employer mobility between survey rounds (two-year intervals) as a lower bound for λ_1^j . In the model, λ_1^j represents the hazard rate of receiving a higher-paying job offer; hence, the proportion of workers experiencing substantial wage increases across waves provides an empirical upper bound. As detailed in [Appendix D](#), these empirical bounds range from approximately 0.4 to 1.3, which aligns well with the model’s estimated λ_1^j values.

4.5 Estimation for 2014

After the 2013 relaxation of China’s One-Child Policy, the gender wage gap in the high-skilled labor market widened by 15.3%, as documented in the empirical analysis. To examine how employers and labor-market dynamics responded to this policy shift so that we observe such a wage change in average, I re-estimate the model for 2014 while remaining agnostic about the specific mechanisms behind the wage adjustment.

Since actual birth rates did not respond immediately to the policy relaxation, I assume that the unemployed benefits b^j remain unchanged between 2012 and 2014. This reflects the absence of any contemporaneous labor-supply adjustment that would alter households’ valuation of non-working time. Moreover, because fertility behavior did not change in the short run, I also assume that workers did not adjust their family-work allocation in anticipation of future births. As discussed in Section 4.2, the exogenous separation rates δ^j are interpreted as demographic “age-out” hazards, which are essential for identifying other parameters in equilibrium. Given that the age distribution of workers remained stable between 2012 and 2014, I hold δ^j fixed at their 2012 values. Similarly, the parameters governing the mapping from family-work time to effort and from effort to human-capital accumulation are held constant.

Once the time spent on family work for individuals with two children is specified, their corresponding effort and accumulation levels can be inferred directly. Knowing the human-capital accumulation rate for workers with two children ($n = 3$) allows us to identify the probability that employers assign to a worker being in that state, $p^j(n = 3)$. In other words, as discussed in the identification argument, what we observe empirically is an “average” accumulation rate—an expectation over possible fertility states. To separate the accumulation rate $\alpha^j(n)$ for each n , we need to know the probability weights $p^j(n)$ that employers attach to each state, and vice versa. Family-work time determines $\alpha^j(n)$, so variation in expected family-work time across fertility states corresponds to variation in employers’ perceived probability that a worker will have two children.

However, the time-use data contain very few observations of individuals with two children. Even among those who gave birth to a second child after the policy change, selection bias is

likely: only women with high endowments or family-friendly jobs could afford to have another child and to reduce their labor supply substantially. Their reported family-work time may therefore overstate the amount that typical employers expect for an average worker with two children. To address this limitation, I impose the following assumption.

Assumption 4.1 (Time use with a second child). For women, the increase in family-work time from one to two children equals 25% of the increase from zero to one child; for men, family-work time remains unchanged. Formally,

$$t^F(3) = t^F(2) + 0.25 [t^F(2) - t^F(1)], \quad t^M(3) = t^M(2).$$

This assumption reflects the diminishing marginal time cost of additional children commonly documented in time-use studies. Using the American Time Use Survey, [Price \(2008\)](#) report that mothers' weekly family-work time rises by about 27% when moving from one to two children, placing the 25% increase well within the empirical range. Holding men's time fixed is consistent with the empirical results in Section 2: women's wages decline significantly following the policy shift, while men's wages remain unaffected—implying that employers did not revise expectations about men's future productivity. Using the reproduced baseline time-use moments $t^F(0) = 0.54$, $t^F(1) = 1.23$, $t^F(2) = 2.60$; $t^M(0) = 0.41$, $t^M(1) = 0.73$, and $t^M(2) = 1.32$ (hours per weekday), the implied family-work times for two children are $t^F(3) \approx 2.943$ and $t^M(3) \approx 1.32$.

Although time-use data from the 2018 CFPS—the first post-reform wave containing the time-use module—show that mothers with two children spend roughly four hours per day on family work, this figure likely reflects realized behavior among a selective group of women rather than employers' expectations about average workers. Nonetheless, because this value may still approximate the true population mean, I consider several counterfactual levels of $t^F(3)$ ranging from 2.995 to 4.0 hours in the estimation. For each assumed $t^F(3)$, employers' perceived probability of a second child, $p_{2014}^F(3)$, adjusts endogenously through the model. By contrast, I do not estimate $p_{2014}^M(3)$, as men's wages show no significant response to changes in expected fertility, implying that $\alpha^M(3)$ and $\alpha^M(2)$ are effectively indistinguishable.

Furthermore, because the first-birth margin remained largely unaffected by the policy change, I impose the following restriction.

Assumption 4.2 (First-birth margin fixed). The relaxation of the OCP does not alter first-birth decisions within the sample window. Only the conditional probability of having a second child changes, implying that $p^j(n = 0)$ and $p^j(n = 1)$ remain the same as in 2012, while $p^j(n = 2)$ and $p^j(n = 3)$ adjust.

Using the 2012 estimates as a baseline, I re-estimate the 2014 model

$$(z_{0,2014}, \lambda_{0,2014}^F, \lambda_{1,2014}^F, \varepsilon^F(3), p_{2014}^F(3), \lambda_{0,2014}^M, \lambda_{1,2014}^M)$$

by minimizing the Anderson–Darling distance between the model-implied and empirical wage distributions for 2014.

Table 14: Estimation Results for 2014 under Alternative t_3 Assumptions

$t^F(3)$	$z_{0,2014}$	λ_0^F	λ_1^F	λ_0^M	λ_1^M	$p_{2014,3}^F$	Fitness
2.955	25,963	0.2695	0.2690	0.2779	0.2445	0.509	975.16
3.450	26,542	0.2590	0.2464	0.2646	0.2551	0.384	894.58
3.750	26,537	0.2656	0.2536	0.2657	0.2563	0.324	896.62
4.000	26,538	0.2664	0.2520	0.2655	0.2561	0.297	902.33

Notes: This table reports the estimated parameters for 2014 under different assumptions about women’s daily family-work time with two children (t_3 , in hours). Each specification re-estimates the common productivity level ($z_{0,2014}$), female and male offer arrival rates while unemployed (λ_0^F, λ_0^M) and while employed (λ_1^F, λ_1^M), and the perceived probability that women have a second child, $p_{2014}^F(3)$. The fitness statistic corresponds to the minimized Anderson–Darling distance; lower values indicate better model fit.

Figure 7b presents the model’s fit to the empirical wage CDF for 2014 (for cases where $t^F(3) = 3.45$), while Table 14 summarizes the corresponding parameter estimates under alternative assumptions about women’s family-work time with two children, $t^F(3)$. Several findings emerge from the estimation.

First, across all specifications, the estimated probability that employers assign to women having a second child, $p_{2014,3}^F$, is strictly positive and quantitatively large. As expected, higher values of $t^F(3)$ —corresponding to greater time devoted to household responsibilities and lower work effort—lead to smaller estimated $p_{2014,3}^F$. Even under the most time-intensive scenario ($t^F(3) = 4$ hours per weekday), which implies that the second child doubles the incremental family-work burden induced by the first child ($4 \approx 2.61 + (2.61 - 1.23)$), the model still yields a sizable $p_{2014,3}^F = 0.297$. This means that among the 51% of women expected to become mothers, employers anticipate that roughly 58% will have a second child following the policy

relaxation.

Second, the results reveal a substantial “productivity depreciation” effect. When the decline in women’s human-capital accumulation rate at $n = 3$ is not large enough, the model fails to match the empirical wage distribution unless the probability of a second child reaches its maximum feasible value ($p_{2014}^F(3) \approx 0.51$), implying that all women would need to have a second child to fit the data—an implausible scenario. Allowing for a sharper decline in productivity therefore improves the model fit dramatically, suggesting that employers expect a significant reduction in women’s future work effort once they have two children.

Third, consistent with the identification argument, the estimates show that $t^F(3)$ and $p_{2014}^F(3)$ cannot be separately identified: different combinations of these two variables yield similar fits, while other key labor-market parameters remain stable across specifications. Only when $t^F(3)$ is set low does the model exhibit poor fit and unstable parameter estimates.

Finally, the estimated initial productivity level, $z_{0,2014}$, rises relative to 2012, capturing aggregate economic growth and reflecting the general upward shift in men’s wages over the same period. This suggests that, overall wage levels in the high-skilled labor market increased with macroeconomic expansion, however, depreciation of female productivity growth due to higher expected family-work burdens made women’s wage decrease.

Discussion. Treating the 2014 wage distribution as a new steady state is admittedly a strong and perhaps unrealistic assumption. In reality, firms cannot instantaneously adjust their posted wage offers within a single year, and given the high degree of labor-market immobility in China, most workers employed in 2012 would have remained in their original positions. Consequently, only new labor-market entrants would have been immediately affected by the policy relaxation.

To account for this, I experimented with a simulation-based estimation approach. In that exercise, I drew incumbents from the 2012 wage distribution, allowed them to receive new job offers from the post-reform offer distribution characterized by the updated λ_1^j , and simulated new entrants drawing offers from $F_{2014}(w)$ also with the updated arrival rate λ_0^j . I then combined these two groups to match the observed 2014 wage distribution. However, this approach also implicitly assumes that all employers instantly re-optimize wage posting

policies and start to post wages based on this new policy within a year. The resulting wage-offer distribution implied by this procedure turned out to be implausible—women’s offers concentrated near the reservation wage, while men’s offers are concentrated where we observe the rightward shifts. This pattern suggests it is more consistent with a transitory adjustment period than with a true post-policy steady state, and if this is not a steady state, using the guessed parameters to formulate an equilibrium wage offer distribution is ill-founded. Given there is no good way to capture the transition path, I return to the more standard approach adopted in the main analysis, estimating the 2014 distribution directly as if it reflects the new equilibrium results while not keeping part of the people fixed in their original distribution.

Despite the steady-state simplification, the estimation results convey several important insights. Empirically, men’s wage distribution shifted upward in the upper quantiles, while women’s distribution moved downward in the lower quantiles. In the canonical search-and-matching framework, the only way to generate a rise in male wages without altering the unemployment margin is through an increase in the initial productivity level z_0 . Yet there is little reason to believe that only men’s initial human capital increased after the reform, as men and women exhibited similar educational attainment and labor-market entry characteristics. Once we allow z_0 to rise for both genders, the observed decline in women’s wages, absent changes in unemployment, must be attributed to slower human-capital accumulation among women.

This interpretation aligns with the model’s mechanism: following the policy relaxation, employers anticipate reduced effective work effort among women. As a result, they discount women’s expected productivity growth more heavily in wage offers, widening the gender wage gap. Hence, the 2014 estimates reinforce the paper’s central argument that the post-reform increase in gender wage inequality stems primarily from expectation-driven differences in human-capital accumulation rates.

As a further check, I overlay the simulated wage CDFs for 2012 and 2014 to compare relative movements with the empirical CDFs in Figure 4. The simulation overlay (Figure 8) mirrors the data: the female CDF shifts left in the lower quantiles, while the male CDF shifts right in the upper quantiles.

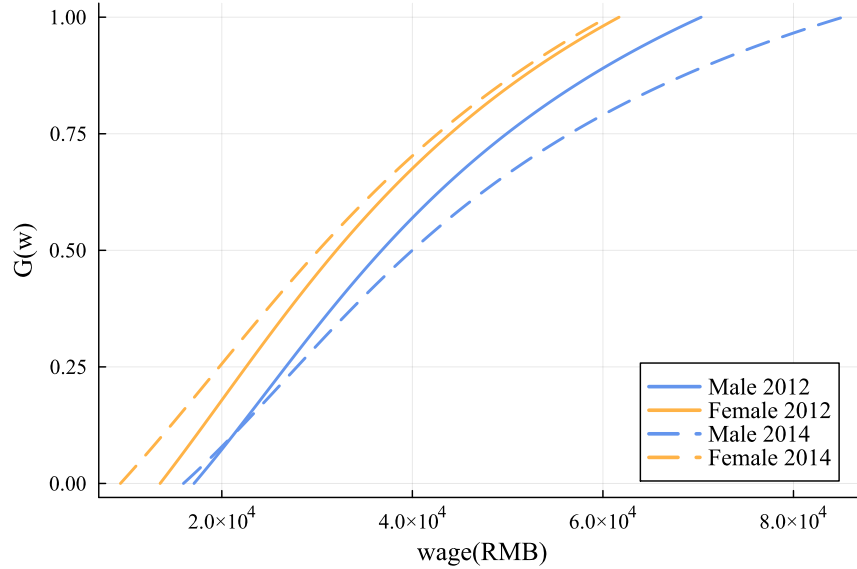


Figure 8: Simulated CDFs (2012 vs. 2014) by Gender: Overlay

4.6 Counterfactual Analysis

In this subsection, I use the estimated parameters to perform a series of counterfactual analyses designed to assess the quantitative importance of gender differences in human-capital accumulation. Specifically, I evaluate how much these differences contribute to (i) the baseline gender wage gap prior to the relaxation of the One-Child Policy, and (ii) the subsequent widening of the gap following the policy change. The exercises decompose the observed wage disparities into components attributable to differences in household time allocation, fertility distribution, and labor-market dynamics, thereby illustrating the central role of human-capital accumulation in shaping gendered wage outcomes both before and after the reform.

Using the 2012 estimates, I first assess how much of the residual gender wage gap—13.3% after controlling for education, industry, and other observables—is driven by unequal family-work time across genders. I construct a counterfactual in which men and women are assigned the same average family-work hours in 2012, holding all other parameters fixed, and solve the model to compute the implied equilibrium wages. Panel A of Table 15 shows that equalizing family-work hours would eliminate the gender wage gap entirely and even slightly reverse it. In contrast, differences in labor-market frictions—such as offer-arrival rates—play a negligible role in explaining the remaining wage differential once observables are controlled for. These

results underscore that the unbalanced division of household labor accounts for the majority of the pre-reform gender wage gap.

Empirically (Section 2), women’s wages declined by 15.3% after the policy relaxation. The model replicates this magnitude closely: the simulated female-to-male wage ratio falls from 0.867 in 2012 to 0.732 in 2014 (Table 15), corresponding to a 13.5-percentage-point decline—or roughly 15.5% if men’s wages are held fixed. To isolate the role of employer expectations, I conduct a second counterfactual in which only the fertility distribution p_n^j is updated to its post-reform values while all other parameters remain at their 2012 levels. The resulting wage ratio of 0.761 implies that nearly 80% of the observed widening in the gender wage gap can be attributed to the expectation channel—employers discounting women’s wages due to higher anticipated fertility and lower expected productivity growth.

Table 15: Gender Wage Ratio Decomposition and Comparison, 2012–2014

Scenario	Female/Male ratio	Gender gap (pp)
<i>Panel A: 2012 Baseline Counterfactuals</i>		
Baseline (2012)	0.867	13.3
Female with male family–work hours	1.021	−0.02
Male with female family–work hours	0.987	0.01
Female with male fertility distribution	0.901	9.9
Female with male labor market parameters	0.889	11.1
<i>Panel B: 2012 vs. 2014</i>		
Baseline (2014)	0.732	26.8
2012 parameters with 2014 productivity	0.761	23.8

Notes: Gender wage gap is defined as $1 - \text{ratio}$ and reported in percentage points (pp). Panel A presents counterfactual exercises based on the 2012 estimated parameters. Panel B compares outcomes between 2012 and 2014 under productivity and distributional shifts. For 2014, results are computed using $t^F(3) = 3.45$, the assumed family–work intensity for women with two children that yields the best model fit.

4.7 Model Implications: Job Protection Policies

The estimation results offer insight into how job protection policies may affect gender wage inequality. Such policies aim to preserve employment continuity after childbirth by requiring firms to retain women’s positions during maternity or childcare leave. Since the child penalty is a key driver of the remaining gender wage gap, and women’s weaker post-birth labor-market attachment limits career progression and human-capital accumulation, job protection is often

viewed as a remedy. However, by committing firms to preserve positions, these policies also extend the non-adjustment horizon over which pay is effectively fixed within a position. Employers must therefore price expected average productivity over the entire match duration, not just contemporaneous output. While job protection secures women’s employment, it can also intensify statistical discrimination at hiring and promotion, leading to lower initial wage offers for all women.

China’s high-skilled labor market provides an illustrative case of an extreme form of job protection—lifetime job security. The results are striking. First, gender wage gaps persist despite continuous female employment. Because firms cannot dismiss workers, employers are not concerned about replacement costs when women have children; instead, they anticipate that longer-term employees facing heavier family responsibilities will divert effort toward childcare while remaining employed at guaranteed pay. As a result, firms discount women’s expected productivity over the contract horizon, lowering their starting wages and promotion probabilities.

Second, job protection effectively keeps both high-effort and low-effort workers in the labor market. This raises the weight of lower-productivity matches in the average expected productivity used to set wages. In an extreme counterfactual, if employers knew that women with two children would leave the labor market, long-term job security would not reduce young women’s wages *ex ante*, because the expected productivity of the employed pool would not decline.

Third, the results underscore that the “baby effect” on women’s wages extends far beyond the short maternity-leave period. Because caregiving responsibilities persist throughout a child’s upbringing, employers price lower productivity over a long horizon, distinguishing a *fertility shock* from a mere *maternity-leave shock*. If the latter were all that mattered, longer job contracts would dilute the penalty rather than magnify it.

Moreover, position-preserving policies can inadvertently heighten exposure to persistent shocks. By prolonging match duration, they increase the likelihood that workers encounter life events—such as childbirth—within the protected employment spell. Empirically, low-educated women are barely affected by the 2013 policy relaxation precisely because they lack long-term contracts; when employers expect short employment spells, the probability of

childbirth during that spell is small, and hence no wage discount is applied.

Consequently, job protection alone may fail to close the gender wage gap and can even shift discrimination to the hiring and promotion margins. The Chinese case—where high-skilled workers enjoy near-perfect job security yet a substantial gender wage gap persists and a large wage drop after the increase of fertility rates (by at most one)—illustrates this paradox. Protecting jobs without addressing the root causes—unequal household specialization and employers’ beliefs about women’s future productivity—leaves wage inequality largely intact. This interpretation aligns with [Goldin \(2021\)](#), who emphasize that the earnings gap reflects career gaps arising from caregiving burdens and the high returns to continuous, long-hour work. Policies that ease childcare constraints and redistribute family responsibilities are therefore more likely to mitigate the gender wage gap than those that simply extend job protection.

5 Conclusion

This paper demonstrates that when pay is contracted in a long-horizon employment relationship, employers’ expectations about workers’ future caregiving burdens are reflected in wages in ways that can materially widen gender gaps. Exploiting China’s 2013 selective relaxation of the One-Child Policy as a quasi-experiment, I document a 15.3 percent decline in young women’s wages with no short-run increase in births, consistent with forward-looking wage setting rather than contemporaneous labor-supply changes. I then develop a search and match model, but extend it to allow for increasing productivity over the match and augment it with a household time-allocation block: effort depends on free time, human capital accumulates with effort, and wages are posted against the present value of an increasing productivity path over the match. Estimating the model on Chinese data, I find that the unequal division of family work alone explains essentially the entire baseline (2012) gender wage gap, and that more than 80 percent of the post-reform widening can be attributed to employers’ downward revisions of women’s expected productivity associated with anticipated second births.

Two broader implications follow. First, in contract-driven pay settings—government and

state-affiliated employment, and many high-skill jobs where matching is costly, training is firm-specific, and replacement is expensive—wages tilt toward expectations about future productivity rather than current output. Any belief that women will accumulate human capital more slowly is amplified by the longer wage-contract horizon and shows up as larger up-front markdowns. Second, policies that merely preserve positions (e.g., job-protection mandates) lengthen the horizon over which present-value pricing applies and can therefore intensify ex-ante markdowns and stall promotions, even as employment is maintained. By contrast, policies that reduce childcare burdens—expanding affordable childcare, designing leave that equalizes caregiving across parents like Sweden’s “second dad paternity leave”—directly raise expected effort and accumulation and are more likely to narrow gaps.

The analysis has limitations. Time-use inputs for family work are observed, but the second-child increment is calibrated due to the absence of a clear within-window birth response; employer expectations about second births therefore rest on an assumption and may be imprecise. On the modeling side, fertility is treated as exogenous rather than the outcome of a household decision. A natural next step is to link the household and labor-market blocks more tightly to study feedback: if employers statistically discriminate against women, lower expected returns to market work may induce higher fertility or reduced effort, which in turn validates firms’ priors. Such a reinforcing cycle could help explain why, under rising childcare burdens, women disproportionately cut effort while men do not.

Overall, the evidence and the model point to a common conclusion: when salaries are predetermined and productivity grows on the job, expectation-driven pricing is an important force behind gender wage gaps. Closing those gaps requires moving the lower expectations that firms legitimately price for women, rather than only lengthening their job tenure.

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A Full Empirical Regression Tables

This appendix reports the full regression tables corresponding to the compact empirical tables in the main text.

Table 16: Effect of OCP Relaxation on Older Workers' Wages (Ages 35-54)

	Panel A: Binary Treatment			Panel B: Cont. Treatment		
	Female	Male	Triple-DiD	Female	Male	Triple-DiD
Dependent variable: log wage						
Years of schooling	0.161*** (0.009)	0.127*** (0.013)	0.143*** (0.009)	0.160*** (0.009)	0.127*** (0.013)	0.143*** (0.009)
Age	0.053 (0.083)	-0.049 (0.065)	-0.023 (0.061)	0.057 (0.084)	-0.050 (0.066)	-0.025 (0.061)
Age ²	-0.001 (0.001)	0.000 (0.001)	0.000 (0.001)	-0.001 (0.001)	0.001 (0.001)	0.000 (0.001)
Treat	-0.063 (0.120)	-0.230* (0.114)				
Treat $\times$ female			0.168 (0.126)			
Treat (cont.)				-0.006 (0.055)	-0.062** (0.028)	
Treat (cont.) $\times$ female						0.053 (0.057)
Observations	1312	1690	3002	1312	1690	3002
R^2	0.211	0.203	0.235	0.210	0.201	0.235

Notes: Province-year panel; standard errors in parentheses. “Treat” is the post-OCP relaxation indicator interacted with above-median pre-policy fine rates; “Treat (cont.)” uses the continuous fine measure. “Triple-DiD” includes interactions with the female indicator and the full set of province $\times$ year, year $\times$ gender, and province $\times$ gender fixed effects.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 17: Full Regression Results: Effect of OCP Relaxation on Female/Male Workers' Wages

	(1) Female	(2) Male	(3) Female	(4) Male	(5) (6) Triple-DiD	
Dependent variable: log wage						
Years of schooling	0.089*** (0.016)	0.067*** (0.022)	0.089*** (0.016)	0.067*** (0.022)	0.078*** (0.009)	0.078*** (0.010)
Age	0.432*** (0.101)	0.501*** (0.066)	0.431*** (0.101)	0.501*** (0.066)	0.455*** (0.072)	0.455*** (0.072)
Age ²	-0.007*** (0.002)	-0.008*** (0.001)	-0.007*** (0.002)	-0.008*** (0.001)	-0.007*** (0.001)	-0.007*** (0.001)
Treat	-0.153* (0.083)	-0.031 (0.084)				
Treat (cont.)			-0.059* (0.031)	0.006 (0.025)		
Treat $\times$ female					-0.148* (0.078)	
Treat (cont.) $\times$ female						-0.075** (0.030)
Observations	1467	1519	1467	1519	2986	2986
R^2	0.200	0.216	0.200	0.215	0.239	0.239

Notes: Columns (1)–(4) include year and province fixed effects. Columns (5)–(6) include province-by-year, year-by-gender, and province-by-gender fixed effects. Standard errors are clustered at the province level.

Table 18: Full Regression Results: Effect of OCP Relaxation on Female Workers' Employment Status

	Panel A: Binary Treatment			Panel B: Cont. Treatment		
	Female	Male	Triple-DiD	Female	Male	Triple-DiD
Dependent variable: waged employment						
Years of schooling	0.173*** (0.018)	0.122*** (0.017)	0.150*** (0.012)	0.174*** (0.018)	0.122*** (0.017)	0.150*** (0.012)
Age	-0.032 (0.057)	0.152* (0.086)	0.058 (0.060)	-0.034 (0.058)	0.157* (0.086)	0.057 (0.060)
Age ²	0.001 (0.001)	-0.002 (0.002)	-0.001 (0.001)	0.001 (0.001)	-0.002 (0.002)	-0.001 (0.001)
Treat	0.115 (0.125)	-0.030 (0.140)				
Treat $\times$ female			0.147 (0.220)			
Treat (cont.)				0.026 (0.049)	0.063 (0.042)	
Treat (cont.) $\times$ female						-0.035 (0.075)
Observations	2241	2068	4296	2241	2068	4296

Notes: Probit specifications. Columns (1), (2), (4), and (5) include year and province fixed effects. Columns (3) and (6) include province-by-year, year-by-gender, and province-by-gender fixed effects. Standard errors are clustered at the province level. Probit tables do not report OLS R^2 .

Table 19: Full Regression Results: Impact of OCP Relaxation on Birth Rates

	(1) Binary	(2) Continuous	(3) Binary	(4) Continuous
Dependent variable: birth rate (%)				
Time=1	0.010 (0.051)	0.016 (0.094)		
Treat=1	0.027 (0.056)			
Time=1 $\times$ Treat=1	0.030 (0.072)			
fines_2000		0.022 (0.027)		
Time=1 $\times$ fines_2000		0.003 (0.034)		
Treat			0.030 (0.021)	
Treat (cont.)				0.003 (0.008)
Observations	216	216	216	216
R^2	0.011	0.018	0.928	0.928

Notes: Columns (1)–(2) exclude province and year fixed effects; Columns (3)–(4) include province and year fixed effects. Robust standard errors are reported.

Table 20: Full Regression Results: Senior Women as an Additional Control Group

	Senior female (1)	Young female (2)	Triple-DiD (3)
Dependent variable: log wage			
Treat (cont.)	-0.019 (0.047)	-0.059* (0.031)	
Treat (cont.) $\times$ young			-0.047* (0.025)
Observations	1252	1467	2719
R^2	0.201	0.200	0.207

Notes: Columns (1)–(2) include year and province fixed effects. Column (3) includes province-by-year fixed effects, year fixed effects, and province-by-age-group fixed effects. Standard errors are clustered at the province level.

Table 21: Full Regression Results: Wage Regressions by Government-Related Sector

	Panel A: Government-related			Panel B: Non-government-related				
	Female	(2)	Triple-DiD	Female	(5)	(6)	Triple-DiD	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable: log wage								
Years of schooling	0.035	0.037	0.051*	0.052*	0.093***	0.093***	0.091***	0.091***
	(0.030)	(0.031)	(0.028)	(0.029)	(0.021)	(0.021)	(0.010)	(0.010)
Age	0.117	0.107	0.127	0.123	0.522***	0.522***	0.518***	0.517***
	(0.104)	(0.107)	(0.112)	(0.108)	(0.120)	(0.120)	(0.076)	(0.076)
Age ²	-0.001	-0.001	-0.002	-0.002	-0.009***	-0.009***	-0.008***	-0.008***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)	(0.001)
Treat	-0.191				-0.146			
	(0.151)				(0.100)			
Treat (cont.)		-0.138***				-0.049		
		(0.045)				(0.040)		
Treat × female			-0.277				-0.195	
			(0.214)				(0.122)	
Treat (cont.) × female				-0.206***				-0.068
				(0.061)				(0.042)
Observations	419	419	861	861	1048	1048	2125	2125
R ²	0.250	0.255	0.311	0.314	0.223	0.223	0.267	0.267

Notes: All columns include year, province, and industry fixed effects. Standard errors are clustered at the province level. Panel A restricts the sample to government-related sectors; Panel B restricts the sample to non-government-related sectors.

Table 22: Full Regression Results: Effect of OCP Relaxation on Female Workers' Promotions

	Has subordinates		Log subordinates	
	(1)	(2)	(3)	(4)
Female x Treat	0.163 (0.264)		-0.017 (0.400)	
Female x Treat (cont.)		0.096 (0.142)		-0.058 (0.152)
Sample mean	0.20	0.20	1.70	1.70
Mean subordinates			11.92	11.92
R^2			0.208	0.208
Observations	2,003	2,003	514	514

Notes: Columns (1)–(2) are probit specifications for having subordinates; Columns (3)–(4) are OLS specifications for log number of subordinates. Specifications include province-by-year, year-by-gender, and province-by-gender fixed effects. Standard errors are clustered at the province level. R^2 is reported only for the OLS columns.

Table 23: Full Regression Results: Effect of OCP Relaxation on Female Wages by Age Group

	Panel A: Binary Treatment			Panel B: Cont. Treatment		
	All ages	Age 29–35	Age 18–28	All ages	Age 29–35	Age 18–28
Dependent variable: log wage						
Years of schooling	0.078*** (0.009)	0.097*** (0.018)	0.060*** (0.012)	0.078*** (0.010)	0.097*** (0.017)	0.060*** (0.012)
Age	0.455*** (0.072)	0.312 (0.348)	0.690*** (0.228)	0.455*** (0.072)	0.311 (0.350)	0.686*** (0.229)
Age ²	-0.007*** (0.001)	-0.005 (0.006)	-0.012** (0.005)	-0.007*** (0.001)	-0.005 (0.006)	-0.012** (0.005)
Treat × female	-0.148* (0.078)	0.052 (0.108)	-0.363** (0.162)			
Treat (cont.) × female				-0.075** (0.030)	0.008 (0.032)	-0.162* (0.084)
Observations	2986	1503	1483	2986	1503	1483
R^2	0.239	0.278	0.232	0.239	0.278	0.232

Notes: Specifications include province-by-year, year-by-gender, and province-by-gender fixed effects. Standard errors are clustered at the province level.

Table 24: Full Regression Results: Effect of OCP Relaxation on Wages, Low-Education Sample

	Panel A: Binary Treatment			Panel B: Cont. Treatment		
	Female	Male	Triple-DiD	Female	Male	Triple-DiD
Dependent variable: log wage						
Years of schooling	0.048* (0.024)	0.074** (0.027)	0.059*** (0.021)	0.048* (0.025)	0.074** (0.027)	0.059*** (0.021)
Age	0.090 (0.119)	0.406*** (0.089)	0.244*** (0.086)	0.089 (0.118)	0.406*** (0.089)	0.244*** (0.086)
Age ²	-0.002 (0.002)	-0.007*** (0.002)	-0.004** (0.002)	-0.001 (0.002)	-0.007*** (0.002)	-0.004** (0.002)
Treat	-0.067 (0.246)	0.171 (0.147)				
Treat $\times$ female			-0.264 (0.251)			
Treat (cont.)				-0.034 (0.115)	0.119*** (0.037)	
Treat (cont.) $\times$ female						-0.140 (0.094)
Observations	511	596	1107	511	596	1107
R^2	0.158	0.142	0.245	0.158	0.146	0.246

Notes: Columns (1), (2), (4), and (5) include year and province fixed effects. Columns (3) and (6) include province-by-year, year-by-gender, and province-by-gender fixed effects. Standard errors are clustered at the province level.

B Derivation and Proofs

B.1 Employment Stock at a Posted Wage

Fix gender j and suppress the superscript. Let $F(w)$ be the wage-offer distribution, $G(w)$ the accepted-wage distribution among employed workers, u the mass of unemployed workers, and $m - u$ the mass of employed workers. The separation hazard at wage w is

$$h(w) = \delta + \lambda_1[1 - F(w)].$$

For $w \geq R$, steady-state flow balance for workers employed at wages no greater than w implies

$$G(w) = \frac{\delta[F(w) - F(R)]}{h(w)[1 - F(R)]}.$$

For any small interval $(w - \epsilon, w]$, the average employment stock per firm among firms posting wages in that interval is

$$(m - u) \frac{G(w) - G(w - \epsilon)}{F(w) - F(w - \epsilon)}.$$

Define the stock of a representative firm posting w as the limiting local average:

$$l(w) = (m - u) \lim_{\epsilon \downarrow 0} \frac{G(w) - G(w - \epsilon)}{F(w) - F(w - \epsilon)}.$$

Substituting the expression for $G(w)$ gives

$$\begin{aligned} l(w) &= (m - u) \delta \lim_{\epsilon \downarrow 0} \frac{\frac{F(w) - F(R)}{h(w)[1 - F(R)]} - \frac{F(w - \epsilon) - F(R)}{h(w - \epsilon)[1 - F(R)]}}{F(w) - F(w - \epsilon)} \\ &= (m - u) \delta \lim_{\epsilon \downarrow 0} \frac{F(w)h(w - \epsilon) - F(w - \epsilon)h(w) + F(R)[h(w) - h(w - \epsilon)]}{h(w)h(w - \epsilon)[1 - F(R)][F(w) - F(w - \epsilon)]}. \end{aligned}$$

Using

$$h(w) - h(w - \epsilon) = -\lambda_1[F(w) - F(w - \epsilon)],$$

the numerator simplifies to

$$[F(w) - F(w - \epsilon)] \{h(w - \epsilon) + \lambda_1[F(w - \epsilon) - F(R)]\}.$$

Therefore

$$l(w) = (m - u)\delta \lim_{\epsilon \downarrow 0} \frac{h(w - \epsilon) + \lambda_1[F(w - \epsilon) - F(R)]}{h(w)h(w - \epsilon)[1 - F(R)]}.$$

Using the aggregate employment flow-balance condition

$$\lambda_0[1 - F(R)]u = \delta(m - u),$$

$$l(w) = \lambda_0 u \lim_{\epsilon \downarrow 0} \frac{h(w - \epsilon) + \lambda_1[F(w - \epsilon) - F(R)]}{h(w)h(w - \epsilon)}.$$

Since $h(w - \epsilon) = \delta + \lambda_1[1 - F(w - \epsilon)]$, the numerator is

$$\delta + \lambda_1[1 - F(R)].$$

Thus

$$l(w) = \lambda_0 u \lim_{\epsilon \downarrow 0} \frac{\delta + \lambda_1[1 - F(R)]}{h(w)h(w - \epsilon)}.$$

Using the no-mass-point property proved in the next subsection, F is continuous at interior support points and $h(w - \epsilon) \rightarrow h(w)$. Hence

$$l(w) = \frac{\lambda_0 u \{\delta + \lambda_1[1 - F(R)]\}}{[h(w)]^2}.$$

Finally, no firm posts below the reservation wage in equilibrium, so the support starts at R and $F(R) = 0$. Hence

$$l(w) = \frac{(\delta + \lambda_1)\lambda_0 u}{[h(w)]^2}.$$

This is the standard Burdett–Mortensen stock-retention formula.

B.2 No Interior Mass Points in the Offer Distribution

Lemma B.1 (Lemma 3.4 restated). *In a stationary wage-posting equilibrium, the offer distribution F has no mass points on the interior of its support.*

Proof. The argument follows the standard no-atom logic in Burdett–Mortensen wage-posting models. Suppose, toward a contradiction, that a positive mass of firms posts the same interior wage w . A firm at that wage can deviate to $w + \eta$, with $\eta > 0$ arbitrarily small. This

deviation places the firm strictly above all firms at the atom. Because employed workers accept only strictly higher offers, the deviating firm gains a discrete recruiting advantage over firms that remain at w : it can attract workers currently employed at w , who would not switch to another firm offering the same wage.

The cost of the deviation is only the arbitrarily small wage increase η . All other payoff components vary continuously as $\eta \downarrow 0$. Hence, for sufficiently small η , the discrete gain from moving above the atom dominates the infinitesimal wage cost. The deviation is profitable, contradicting equilibrium profit equalization among wages posted with positive probability. Therefore no interior atom can exist. $\square$

B.3 Average-Productivity Identification

This appendix provides the proof idea behind the average-productivity identification argument in Section 4.2. Fix δ . Given a candidate λ_1 , the observed employed-wage distribution $G(w)$ implies

$$h(w) = \frac{\delta(\delta + \lambda_1)}{\delta + \lambda_1 G(w)}.$$

Let $k = \lambda_1/\delta$. Then

$$h(G) = \frac{\delta(1 + k)}{1 + kG}.$$

Substituting this into Equation (13) gives an observed wage schedule as a rational function of G :

$$w(G) = \frac{\bar{\beta} \delta(1 + k)}{\delta(1 + k) - \bar{\alpha}(1 + kG)} - \frac{1}{(1 + kG)^2} \left[\frac{\bar{\beta}(\delta + \lambda_1)}{\delta + \lambda_1 - \bar{\alpha}} - R \right].$$

This rational function has two relevant denominator terms. The second term has denominator $(1 + kG)^2$, so, when its coefficient is nonzero, it has a double pole at $G = -1/k$. Equality of two such rational functions over an interval therefore requires the same value of k , which identifies $\lambda_1 = k\delta$.

Once k is known, the first term has denominator

$$\delta(1 + k) - \bar{\alpha}(1 + kG).$$

Its pole is located at

$$G = \frac{\delta(1+k) - \bar{\alpha}}{\bar{\alpha}k}.$$

Thus the location of this pole identifies $\bar{\alpha}$. Given k and $\bar{\alpha}$, the coefficient of the first rational term identifies $\bar{\beta}$. Therefore, with fixed δ , the average-productivity wage restriction identifies $(\lambda_1, \bar{\alpha}, \bar{\beta})$, provided the wage support contains nondegenerate variation and the relevant rational terms do not vanish.

B.4 Proof for Identification Theorem

Proof. This subsection provides the logic behind the state-specific identification argument in Section 4.2. Fix (δ, λ_1) and suppose the fertility-state probabilities $\{p_n\}_{n=0}^{\bar{n}}$ are known. The goal is to show how the equilibrium wage schedule identifies the accumulation rates $\alpha(n)$ and the productivity scales $z_0\varepsilon(n)$.

Given an accepted-wage distribution G , the offer distribution and separation hazard are recovered from

$$F(w) = \frac{(\delta + \lambda_1)G(w)}{\delta + \lambda_1 G(w)}, \quad h(w) = \frac{\delta(\delta + \lambda_1)}{\delta + \lambda_1 G(w)}.$$

Equilibrium profit equalization implies that, after eliminating the common profit level using the lower support point R ,

$$w = \frac{h(w)^2}{(\delta + \lambda_1)^2} R + \sum_{n=0}^{\bar{n}} p_n z_0 \varepsilon(n) \left[\frac{h(w)}{h(w) - \alpha(n)} - \frac{h(w)^2}{(\delta + \lambda_1)[\delta + \lambda_1 - \alpha(n)]} \right].$$

Let $x = h(w)$ and $\beta_n = z_0 \varepsilon(n)$. The fertility-state component of the profit identity is

$$T(x) = \sum_{n=0}^{\bar{n}} p_n \beta_n \left[\frac{x}{x - \alpha(n)} - \frac{x^2}{(\delta + \lambda_1)(\delta + \lambda_1 - \alpha(n))} \right].$$

The condition $\alpha(n) < \delta$ guarantees that all denominators are well defined on the equilibrium support. Since $h(w) \geq \delta$, we have $x - \alpha(n) = h(w) - \alpha(n) > 0$. Also, $\delta + \lambda_1 - \alpha(n) > 0$.

For notational simplicity, normalize $\delta + \lambda_1 = 1$. Then

$$T(x) = \sum_{n=0}^{\bar{n}} p_n \beta_n \left[\frac{x}{x - \alpha(n)} - \frac{x^2}{1 - \alpha(n)} \right].$$

Using

$$\frac{x}{x - \alpha} = 1 + \frac{\alpha}{x - \alpha},$$

we can write

$$T(x) = \sum_n p_n \beta_n - x^2 \sum_n \frac{p_n \beta_n}{1 - \alpha(n)} + \sum_{\alpha(n) \neq 0} \frac{p_n \beta_n \alpha(n)}{x - \alpha(n)}. \quad (15)$$

Suppose another parameter vector defines $T'(x)$ and generates the same wage-hazard relationship, so that $T'(x) = T(x)$ over the observed support. Since both sides are well-defined rational functions of x , equality over an interval implies that their partial-fraction representations must coincide. In the representation in Equation 15, each distinct nonzero accumulation rate $\alpha(n)$ appears as the location of a simple-fraction term, $1/[x - \alpha(n)]$. Therefore, there exists a permutation ω over the states with distinct nonzero accumulation rates such that

$$\alpha'(\omega(n)) = \alpha(n).$$

The coefficient on the corresponding simple-fraction term must also coincide:

$$p_{\omega(n)} \beta'_{\omega(n)} \alpha'(\omega(n)) = p_n \beta_n \alpha(n).$$

Using $\alpha'(\omega(n)) = \alpha(n)$, this becomes

$$p_{\omega(n)} \beta'_{\omega(n)} = p_n \beta_n.$$

If p_n is attached to known fertility states and $\alpha(n)$ is distinct across those states, then the matching is pinned down by state labels:

$$\beta'_n = \beta_n.$$

Thus the wage schedule identifies each distinct nonzero accumulation rate $\alpha(n)$ and the corresponding productivity scale $\beta_n = z_0 \varepsilon(n)$, up to relabeling.

If effort is normalized in one reference state n_0 , for example $\varepsilon(n_0) = 1$, then $\beta_{n_0} = z_0$,

which identifies z_0 . The remaining effort levels are then identified as

$$\varepsilon(n) = \frac{\beta_n}{z_0}.$$

□

Caveats. The argument requires distinct nonzero accumulation rates. If two fertility states have the same $\alpha(n)$, the wage schedule identifies only their combined probability-weighted productivity scale. If $\alpha(n) = 0$, that state does not generate a pole, so it requires additional restrictions or normalization to be separately identified.

C Computational Details in Estimation

Objective. Let $G_{\text{emp},g}(w)$ denote the empirical CDF of wages for group $g \in \{m, f\}$ and $G_g(w; \theta)$ the model-implied CDF. In the on-the-job search structure used here, the model CDF is generated from the solved offer CDF $F_g(w; \theta)$ via

$$G_g(w; \theta) = \frac{\delta^g F_g(w; \theta)}{\delta^g + \lambda_1^g (1 - F_g(w; \theta))}. \quad (16)$$

Let $R_g(\theta)$ be the reservation wage and $\bar{w}_g(\theta)$ the upper support obtained from the model's closed-form bound. For each group we minimize a discretized Anderson–Darling (AD) loss that up-weights discrepancies near the CDF tails. On a fine grid $\{w_{g,j}\}_{j=1}^{M_g} \subset [R_g(\theta), \bar{w}_g(\theta)]$ the group-specific criterion is approximated by a trapezoidal sum

$$AD_g(\theta) \approx \sum_{j=1}^{M_g-1} \frac{(\Delta_{g,j}(\theta))^2 + (\Delta_{g,j+1}(\theta))^2}{(\hat{G}_{g,j}(1 - \hat{G}_{g,j}) + \varepsilon) + (\hat{G}_{g,j+1}(1 - \hat{G}_{g,j+1}) + \varepsilon)} \times \frac{w_{g,j+1} - w_{g,j}}{2}, \quad (17)$$

$$\Delta_{g,j}(\theta) := G_g(w_{g,j}; \theta) - \hat{G}_{g,j}, \quad \hat{G}_{g,j} := G_{\text{emp},g}(w_{g,j}),$$

with a small stabilizer $\varepsilon > 0$ to avoid exploding weights when $\hat{G}_{g,j} \in \{0, 1\}$. The sample objective is the sum across groups,

$$AD(\theta) = \sum_{g \in \{m, f\}} AD_g(\theta). \quad (18)$$

The parameter vector θ stacks (i) group-specific reservation-wage components $\alpha^g = (\alpha_0^g, \alpha_1^g, \alpha_2^g)$, (ii) λ_0^g, λ_1^g , and (iii) a common z_0 . Monotonicity and feasibility are enforced by smooth reparameterizations: $\alpha_0^g \geq \alpha_1^g \geq \alpha_2^g < \delta^g$ (via a logistic stick-breaking transform) and $\lambda_0^g > \lambda_1^g$. We obtain $\hat{\theta}$ by multi-start differential evolution and a local Nelder-Mead refinement under these constraints.

Asymptotic normality and standard errors. By Donsker’s theorem, the empirical CDF satisfies $\sqrt{n}(\hat{G}_{\text{emp},g}(\cdot) - G_{0,g}(\cdot)) \Rightarrow \mathbb{G}_g \circ G_{0,g}(\cdot)$, where $\mathbb{G}_g$ is a mean-zero Gaussian bridge. Under standard M -estimation regularity conditions the AD estimator is consistent and asymptotically normal:

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\theta_0)^{-1}).$$

In practice we approximate the (observed) Hessian of the sample AD criterion at the optimum,

$$\hat{H} = \frac{\partial^2 AD(\theta)}{\partial \theta \partial \theta'} \Big|_{\theta=\hat{\theta}}, \quad (19)$$

using finite differences, and we add a small ridge $\lambda_{\text{ridge}}I$ for numerical stability. Our baseline (information-matrix) standard errors are computed from

$$\widehat{\text{Var}}(\hat{\theta}) = (\hat{H} + \lambda_{\text{ridge}}I)^{-1}, \quad \text{SE}(\hat{\theta}_i) = \sqrt{[(\hat{H} + \lambda_{\text{ridge}}I)^{-1}]_{ii}}, \quad (20)$$

with 95% confidence intervals $\hat{\theta}_i \pm 1.96 \text{SE}(\hat{\theta}_i)$.

D Panel Data Results in CFPS Data

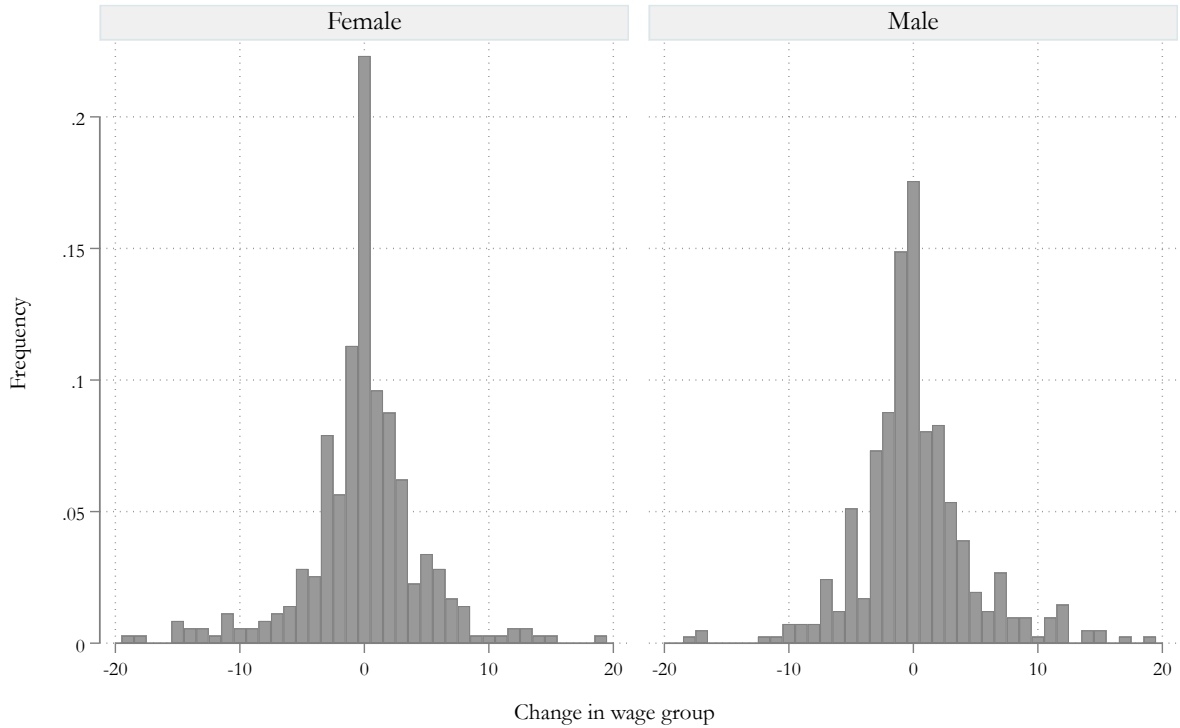
Check with panel data results One way to assess whether the estimated offer-arrival rates are consistent with observed labor-market dynamics in China is to examine how frequently the same individuals experience promotions (wage increases) or job changes across survey waves. Neither of these empirical moments directly corresponds to the promotion probability λ_1^j in the model, since in the model a worker’s wage increases only when a new offer drawn from the offer distribution exceeds the current wage—either from the same employer or through a job-to-job transition. In contrast, in the data, wages may rise within the same position due

to seniority pay, performance bonuses, or cost-of-living adjustments; such increases do not necessarily reflect new offers. Conversely, a job change in the data may result from either involuntary separation or a voluntary job-to-job transition. While the latter corresponds exactly to a promotion in the model, it excludes cases of within-firm promotions.

Therefore, these empirical moments serve only as approximate benchmarks for the estimated λ_1^j . On average, the promotion frequency observed in the data should correspond to the expected frequency of upward moves implied by the model, given by

$$\int_R^{\bar{w}} \lambda_1 (1 - F(w)) dF(w) = \frac{1}{2} \lambda_1. \quad (21)$$

Because we observe only survey waves spaced two years apart, the probability of remaining in the same wage bin over both periods satisfies $(1 - \frac{1}{2} \lambda_1^j)^2 = \text{Pr}(\text{staying})$. This relationship provides a simple way to compare model-implied offer arrival rates with the empirical frequency of observed wage stagnation.



Graphs by gender

Figure 9: Change in Gender-Specific Wage Groups

Because (real) wages are continuous, the probability that an individual's wage remains

exactly the same across two consecutive survey waves is virtually zero. Therefore, I divide the male and female wage distributions into 20 gender-specific bins and interpret a promotion as a wage increase large enough to move an individual from one bin to a higher one. Figure 9 presents the distribution of wage-bin changes between 2012 and 2014 for women and men. For women, among those whose wages did not decline, 12.3% remained in the same wage bin. This implies that the on-the-job offer arrival rate, λ_1 , interpreted as the promotion probability, is approximately $0.649 \times 2 = 1.298$. For men, 11.5% of those whose wages did not decline remained in the same bin, implying an on-the-job offer arrival rate of $\lambda_1 \approx 0.661 \times 2 = 1.322$.

Changes in wage levels provide only an upper bound for identifying new job offers, since wage increases do not necessarily reflect promotions. However, when an employed individual switches employers, it is certain that a new offer has been accepted. Table 25 reports the proportion of individuals who remained with the same employer across two survey waves, corresponding to the probability of staying, i.e., $(1 - \text{Prob}(\text{switching}))^2$.

Among highly educated young workers who remained in the labor force between 2012 and 2014, 44.73% stayed with the same employer. This implies an employer-to-employer transition rate of approximately 0.662 in 2012. Between 2014 and 2016, 67.75% of workers stayed with the same employer, implying a lower transition rate of 0.354 in 2014. Thus, over this period, the implied range of λ_1 —the on-the-job offer arrival rate—lies roughly between 0.35 and 0.66, reflecting the low mobility characteristic of China’s formal labor market for highly educated workers. These empirical moments provide a suggestive lower bound for the estimated λ_1 in the structural model.

Table 25: Job Continuity Across Surveys

	2012 → 2014	(Employed in 2012)
	High education	Non-high education
<i>All ages</i>		
Stayed in same main job	49.12%	43.27%
Not staying	45.96%	48.00%
Out of labor market	4.92 %	8.73%
<i>Age ≤ 35</i>		
Stayed in same main job	44.73%	39.01%
Not staying	50.35%	50.34%
Out of labor market	4.92%	10.65%

Notes: Data are from the 2012 and 2014 CFPS. The sample consists of urban respondents observed in both waves and employed in 2012. High education denotes at least 12 years of schooling; age is measured in 2014. Entries are percentages within each education-by-age subsample and sum to 100, subject to rounding. “Stayed in same main job” requires being employed in 2014 and reporting the same main job as in 2012. “Not staying” includes other observed 2014 labor-market outcomes, while “Out of labor market” denotes respondents coded as outside the labor force in 2014.